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Aircraft with thermal energy storage system

Abstract

A thermal energy system for use with an aircraft includes a cooling loop and a cooler. The cooling loop includes a fluid conduit and a pump configured to move fluid through the fluid conduit to transfer heat from a heat source to the fluid in the fluid conduit to cool the heat source. The cooler includes an air-stream heat exchanger located in a duct and is in thermal communication with the fluid conduit to transfer heat between the fluid in the cooling loop and the air passing through the duct.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/491,470, filed 30 Sep. 2021, which claims priority to and the benefit of U.S. Provisional Patent Application No. 63/086,804, filed 2 Oct. 2020, the disclosures of which are now expressly incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to aircraft, and more specifically to aircraft with thermal energy systems for cooling heat generating components of the aircraft.

BACKGROUND

(2) Gas turbine engines are used to power aircraft, watercraft, power generators, and the like. Gas turbine engines typically include a compressor, a combustor, and a turbine. The compressor compresses air drawn into the engine and delivers high pressure air to the combustor. In the combustor, fuel is mixed with the high pressure air and is ignited. Products of the combustion reaction in the combustor are directed into the turbine where work is extracted to drive the compressor and, sometimes, an output shaft. Left-over products of the combustion are exhausted out of the turbine and may provide thrust in some applications.

(3) Some aircraft may include a hybrid electric propulsion system that may use one or more electric machines, such as generators and motors, to provide power to the aircraft. To reject waste heat produced by the engine, aircraft oil/hydraulic heat loads, power electronics, and/or the electric machines, the aircraft may use a ram-air heat exchanger to reject waste heat to the surrounding air in the atmosphere. However, when the aircraft is grounded or taxiing, the air may be relatively warm relative to the components to be cooled. Moreover, there may be little to no ram air pressure to force the air through the ram-air heat exchanger when the aircraft is grounded or taxiing, limiting the waste heat rejection.

(4) As a result, the ram air duct and heat exchanger may be relatively large to achieve the large mass flow rates to effectively transfer heat to the air in the atmosphere at low ram pressure conditions. Conversely, at cruise altitude, the air temperature is cooler and the ram air pressure difference across the ram-air heat exchanger may have increased due to higher flight speed, which would allow for a smaller ram-air heat exchanger to be used, but for the insufficient cooling rate while grounded and taxiing.

(5) To enable a smaller ram-air heat exchanger when the aircraft is grounded or taxiing, the aircraft may include a blower to draw air through the ram-air heat exchanger, adding weight to the aircraft. As a result, providing a thermal energy system that enables a ram-air heat exchanger configured to maximize fuel efficiency and decrease the added weight to the aircraft presents design challenges.

SUMMARY

(6) The present disclosure may comprise one or more of the following features and combinations thereof.

(7) A thermal energy system for use with an aircraft may include a heat source, a cooling loop, a

cooler, and a thermal-storage fuel system. The cooling loop may have a fluid conduit and a pump configured to move fluid through the fluid conduit to transfer heat from the heat source to the fluid to cool the heat source. The cooler may include a duct configured to conduct air through the duct and an air-stream heat exchanger located in the duct and in thermal communication with the fluid in the cooling loop to transfer heat between the cooling loop and the air conducted through the duct.

(8) In some embodiments, the thermal-storage fuel system may include a first fuel tank and a fuel-tank heat exchanger. The first fuel tank may be configured to store fuel therein. The fuel-tank heat exchanger may be in thermal communication with the fluid and configured to transfer heat between the fluid in the cooling loop and fuel stored in the first fuel tank.

(9) In some embodiments, the cooling loop may be structured to conduct the fluid heated from the heat source first through the air-stream heat exchanger in the duct and then to through the fuel-tank heat exchanger after the air-stream heat exchanger.

(10) In some embodiments, the thermal-storage fuel system may further include a second fuel tank. The second fuel tank may be in fluid communication with the first fuel tank.

(11) In some embodiments, the second fuel tank may also be configured to store a fuel at a temperature different from a temperature of the fuel in the first fuel tank. In some embodiments, the second fuel tank may also be configured to store fuel at a temperature lower than a temperature of the fuel in the first fuel tank.

(12) In some embodiments, the thermal-storage fuel system may further include an engine-fuel unit. The engine-fuel unit may be configured to receive fuel from the second fuel tank and deliver the fuel to an engine.

(13) In some embodiments, the engine-fuel unit may be in thermal communication with the cooling loop to transfer heat between the fluid in the cooling loop and the fuel in the engine-fuel unit.

(14) In some embodiments, the engine-fuel unit may include a valve system and an engine-fuel heat exchanger. The valve system may be in fluid communication with the second fuel tank. The engine-fuel heat exchanger may be in thermal communication with the valve system and the cooling loop to transfer heat between the fluid in the cooling loop and the fuel from the second fuel tank. In some embodiments, the valve system may be configured to vary a flow of fuel through the engine-fuel heat exchanger to deliver fuel to the engine at a predetermined threshold fuel temperature.

(15) In some embodiments, the valve system includes a mix valve, a first conduit, and a second conduit. The first conduit may be in fluid communication between the mix valve and the second fuel tank to deliver fuel having a first temperature to the mix valve. The second conduit may be in fluid communication between the mix valve and the second fuel tank.

(16) In some embodiments, the engine-fuel heat exchanger may be in thermal communication with the second conduit to cause the second conduit to deliver fuel having a second temperature to the mix valve. The mix valve may be configured to vary a first flow rate of fuel from the first conduit and a second flow rate of fuel from the second conduit to provide a mixed stream of fuel.

(17) In some embodiments, the mix stream of fuel may have a third temperature. The third temperature may be less than or equal to the predetermined threshold fuel temperature.

(18) In some embodiments, the cooling loop may be structured to conduct the fluid heated from the heat source first through the engine-fuel heat exchanger, through the air-stream heat exchanger after the engine-fuel heat exchanger, and through the fuel-tank heat exchanger after the air-stream heat exchanger.

(19) In some embodiments, the cooling loop may include a valve. The valve may be connected to the fluid conduit and configured to selectively cause at least a portion of a flow of fluid in the cooling loop to bypass the fuel-tank heat exchanger.

(20) In some embodiments, the cooling loop may include modulating valves. The modulating valves may be connected to the fluid conduit and configured to vary the flow of fluid in the cooling

loop through the air-stream heat exchanger.

(21) In some embodiments, the thermal energy system may further comprise a control system. The control system may be configured to selectively vary a flow of the fluid in the cooling loop through the air-stream heat exchanger and the fuel-tank heat exchanger to maintain a temperature of the heat source below a predetermined heat load temperature. In some embodiments, the control system may be configured to vary the predetermined heat load temperature throughout a flight cycle of the aircraft.

(22) In some embodiments, the control system may be configured to measure the heat transferred between the fluid and the fuel in the first fuel tank. The control system may also be configured to compare the heat transfer measured to a predetermined heat rejection schedule. The control system may also be configured to selectively vary the flow of fluid in the cooling loop through the fuel-tank heat exchanger in response to the heat transfer measured being different from the predetermined heat rejection schedule.

(23) In some embodiments, the control system may be configured to allow the flow of the fluid in the cooling loop through the fuel-tank heat exchanger. The control system may be configured to allow the flow of fluid through the fuel-tank heat exchanger in response to a temperature of the fuel in the first fuel tank being below a first fuel tank predetermined threshold temperature.

(24) In some embodiments, the control system may be configured to block the flow of the fluid in the cooling loop through the fuel-tank heat exchanger in response to a temperature of the fluid in the cooling loop directly upstream of the fuel-tank heat exchanger being above the first fuel tank predetermined threshold temperature while the temperature of the fuel in the first fuel tank is equal to or exceeds the first fuel tank predetermined threshold temperature. The controller may also be configured to allow the flow of the fluid in the cooling loop through the fuel-tank heat exchanger in response to a temperature of the fluid in the cooling loop directly upstream of the fuel-tank heat exchanger being below the first fuel tank predetermined threshold temperature while the temperature of the fuel in the first fuel tank is equal to or exceeds the first fuel tank predetermined threshold temperature.

(25) In some embodiments, the cooler may be a ram air cooler. The duct may be configured to receive air from atmosphere around the aircraft during forward movement of the aircraft relative to ground. In some embodiments, the duct may be free of any air mover.

(26) According to another aspect of the present disclosure, a method may include providing a thermal energy system for use with an aircraft. The thermal energy system may include a cooling loop, a cooler, and a thermal-storage fuel system.

(27) In some embodiments, the cooling loop may have a fluid conduit and a pump. The pump may be configured to move fluid through the fluid conduit.

(28) In some embodiments, the cooler may include a duct and an air-stream heat exchanger. The duct may be configured to conduct air through the duct. The air-stream heat exchanger may be located in the duct and in thermal communication with the fluid in the cooling loop to transfer heat between the cooling loop and the air conducted through the duct.

(29) In some embodiments, the thermal-storage fuel system may include a first fuel tank and fuel-tank heat exchanger. The first fuel tank may be configured to store fuel therein. The fuel-tank heat exchanger may be in thermal communication with the fluid in the cooling loop.

(30) In some embodiments, the method may further include conducting the fluid through the cooling loop to transfer heat between a heat source and the fluid in the cooling loop, conducting the fluid in the cooling loop to the air-stream heat exchanger to transfer heat between the fluid in the cooling loop and the air passing through the duct, and conducting the fluid in the cooling loop to the fuel-tank heat exchanger to transfer heat between the fluid in the cooling loop and fuel stored in the first fuel tank.

(31) In some embodiments, the method may further include selectively varying a flow of the fluid in the cooling loop through the air-stream heat exchanger and a flow of the fluid through the fuel-

tank heat exchanger to maintain a temperature of the heat source below a predetermined heat load temperature.

(32) In some embodiments, the thermal-storage fuel system may further include a second fuel tank. The second fuel tank may be in fluid communication with the first fuel tank. The second fuel tank may be configured to store fuel at a temperature that is different from a temperature of the fuel in the first fuel tank.

(33) In some embodiments, the thermal-storage fuel system may further include an engine-fuel unit. The engine-fuel unit may be configured to receive fuel from the second fuel tank and deliver the fuel to an engine and the engine-fuel unit is in thermal communication with the cooling loop.

(34) In some embodiments, the method may further include conducting the fluid in the cooling loop to the engine-fuel unit before conducting the fluid to the air-stream heat exchanger to transfer heat between the fluid in the cooling loop and the fuel in the engine-fuel unit.

(35) In some embodiments, the engine-fuel unit may include a valve system and an engine-fuel heat exchanger. The valve system may be in fluid communication with the second fuel tank. The engine-fuel heat exchanger may be in thermal communication with the valve system and the cooling loop to transfer heat between the fluid in the cooling loop and the fuel from the second fuel tank. In some embodiments, the valve system may be configured vary a flow of fuel through the engine-fuel heat exchanger to deliver fuel to the engine at an engine-fuel unit predetermined threshold temperature.

(36) In some embodiments, the valve system may include a mix valve, a first conduit, and a second conduit. The first conduit may be in fluid communication between the mix valve and the second fuel tank to deliver fuel having a first temperature to the mix valve. The second conduit may be in fluid communication between the mix valve and the second fuel tank.

(37) In some embodiments, the engine-fuel heat exchanger may be in thermal communication with the second conduit to cause the second conduit to deliver fuel having a second temperature to the mix valve. The mix valve may be configured to vary a first flow rate of fuel from the first conduit and a second flow rate of fuel from the second conduit to provide a mixed stream of fuel.

(38) In some embodiments, the mix stream may have a third temperature. The third temperature may be less than or equal to the engine-fuel unit predetermined threshold temperature.

(39) In some embodiments, the method may further include measuring the heat transfer between the fluid and the fuel in the first fuel tank. The method may further include comparing the heat transfer measured to a predetermined heat rejection schedule. The method may further include selectively varying the flow of fluid in the cooling loop through the fuel-tank heat exchanger in response to the heat transfer measured being different from the predetermined heat rejection schedule.

(40) In some embodiments, the method may further include measuring the temperature of the fluid in the cooling loop upstream of an accumulator included in the cooling loop. The method may further include comparing the temperature measured to a sensor target temperature. The method further includes selectively varying the flow of fluid in the cooling loop through the air-stream heat exchanger in response to the temperature measured being different from the sensor target temperature.

(41) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is elevation view of an aircraft that includes a gas turbine engine and a thermal energy system connected with the gas turbine engine to manage the temperatures of the electric

components of the aircraft and the gas turbine engine;

(2) FIG. 2 is a diagrammatic view of the thermal energy system included in the aircraft of FIG. 1 showing the thermal energy system includes a cooling loop configured to move fluid through system to transfer heat from a heat source to the fluid in the fluid conduit to cool the heat source, a ram air cooler configured to transfer heat between the fluid in the cooling loop to air passing through the ram air cooler, a thermal-storage fuel system configured to transfer heat between the fluid in the cooling loop and fuel stored in a first fuel tank;

(3) FIG. 3 is a diagrammatic view similar to FIG. 2 showing the thermal energy system further includes a controller coupled to a valve connected to the fluid conduit in the cooling loop to selectively cause the flow of fluid to bypass the fuel-tank heat exchanger;

(4) FIG. 4 is a diagrammatic view of another embodiment of a thermal energy system adapted for use in the aircraft of FIG. 1 showing the thermal energy system includes a cooling loop, a ram air cooler, a thermal-storage fuel system configured to transfer heat between the fluid in the cooling loop and fuel stored in a first fuel tank, and an engine-fuel unit configured to transfer heat between the fluid in the cooling loop and fuel from a second fuel tank before being delivered to the gas turbine engine;

(5) FIG. 5 is a diagrammatic view of another embodiment of a thermal energy system adapted for use in the aircraft of FIG. 1 showing the thermal energy system includes a cooling loop configured to move oil through system to transfer heat from a heat source to the oil in the fluid conduit to cool the heat source, a ram air cooler, a thermal-storage fuel system configured to transfer heat between the oil in the cooling loop and fuel stored in a first fuel tank, and an engine-fuel unit configured to transfer heat from the oil in the cooling loop to the fuel stored in a second fuel tank before being delivered to the gas turbine engine;

(6) FIG. 6 is a diagrammatic view of another embodiment of a thermal energy system adapted for use in the aircraft of FIG. 1 showing the thermal energy system includes a cooling loop, a ram air cooler, a thermal-storage fuel system, and a low temperature heat load in thermal communication with the fluid conduit for transferring heat to the fluid in the fluid conduit;

(7) FIG. 7 is a diagrammatic view similar to FIG. 6 showing the thermal energy system further includes a control system having a plurality of valves connected to the fluid conduit and a controller connected to the plurality of valves to operate the plurality of valves and control the flow of fluid through the thermal-storage fuel system; and

(8) FIG. 8 is a diagrammatic view of another embodiment of a thermal energy system adapted for use in the aircraft of FIG. 1 showing the thermal energy system includes a cooling loop, a ram air cooler, a thermal-storage fuel system, and a valve connected to the fluid conduit and configured to selectively cause the flow of fluid to bypass a compressor downstream of a low temperature heat load in thermal communication with the fluid conduit for transferring heat to the fluid in the fluid conduit.

DETAILED DESCRIPTION OF THE DRAWINGS

(9) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

(10) A thermal energy system **12** according to the present disclosure is adapted for use with an aircraft **10** such as the aircraft **10** shown in FIG. 1. The thermal energy system **12** is configured to manage temperatures of components of the aircraft **10**, such as components of a gas turbine engine **14**, electric batteries, motor-generators, and/or oil/hydraulic heat loads **16** of the aircraft **10**.

(11) The thermal energy system **12** includes a cooling loop **20**, a cooler **22**, and a thermal-storage fuel system **24** as shown in FIGS. 2 and 3. The cooling loop **20** has a fluid conduit **30** and a pump **32** configured to move fluid through the fluid conduit **30** to transfer heat from a heat source **16** to the fluid thereby cooling the heat source **16**. The cooler **22** includes a duct **34** configured to conduct air therethrough and a heat exchanger **36** located in the duct **34**. The air-stream heat exchanger **36**

is in thermal communication with the fluid in the cooling loop **20** to transfer heat between the cooling loop **20** and the air conducted through the duct **34**. The thermal-storage fuel system **24** is configured to transfer heat between the fluid in the cooling loop **20** and fuel stored in the thermal-storage fuel system **24**. As such, one or both of the cooler **22** and the thermal-storage fuel system **24** may transfer heat with the fluid.

(12) The thermal-storage fuel system **24** includes a first fuel tank **44**, a second fuel tank **46**, and a fuel-tank heat exchanger **48** as shown in FIGS. **2** and **3**. Both the first fuel tank **44** and second fuel tank **46** are configured to store fuel therein. The second fuel tank **46** is in fluid communication with the first fuel tank **44** through the valves **52**, **53** to transfer fuel selectively therebetween. The fuel-tank heat exchanger **48** is in thermal communication with the fluid conduit **30** and is configured to transfer heat between the fluid in the cooling loop **20** and fuel stored in the first fuel tank **44**. In the illustrative embodiment, the thermal-storage fuel system **24** includes a fuel pump **47** configured to pump fuel through the fuel-tank heat exchanger **48**.

(13) In some embodiments, the cooler **22** may be a ram air cooler **22** and the duct **34** is configured to receive air from the atmosphere **18** around the aircraft **10** during forward movement of the aircraft **10** relative to the ground **19** and a ram-air heat exchanger **36** located in the duct **34**. The duct **34** may be free of any air movers (i.e. blades, fans, blowers, etc.). Aircraft may often use ram air to cool heat loads of the aircraft. Such thermal energy systems rely on ram-air heat exchangers to reject the waste heat produced by power electronic devices or electric machines such as generators, motors, and/or engines. However, the temperature difference between the component being cooled and the temperature of the air flowing through the ram-air heat exchanger may cause design challenges.

(14) For example, when the aircraft **10** is stationary or taxiing, little or no ram pressure is created to force air through the ram-air heat exchanger **36**. As a result, there is little or no air to cool the coolant in the cooling loop **20**. Therefore, some typical aircraft may include a blower to force air through the ram-air heat exchanger, which may add additional weight and volume to the system. The system of the present disclosure does not include or use a blower.

(15) The thermal energy system **12** of the present disclosure includes the thermal-storage fuel system **24** to eliminate the blower and reduce a size of the duct **34** and heat exchanger compared to typical systems thereby increasing efficiency of the thermal energy system **12** and reducing size and weight of the aircraft. The thermal-storage fuel system **24** uses the on-board fuel in the first fuel tank **44** as a heat sink for the heat loads **16** produced early in the flight cycle. In the illustrative embodiment, the cooling loop **20** is structured to conduct the fluid heated from the heat source **16** through the air-stream heat exchanger **36** followed by the fuel-tank heat exchanger **48** after the air-stream heat exchanger **36**.

(16) In this way, the fuel in the first and second fuel tanks **44**, **46** is not transported to the heat loads **16**, but rather, the fluid is brought to the fuel to transfer the heat with the fuel. Transferring heat with the fuel in the first fuel tank **44** minimizes safety risks, and heating the fuel prior to burning in the engine **14** may provide additional benefits.

(17) In the illustrative embodiment, the thermal energy system **12** further includes a control system **26** as shown in FIGS. **2** and **3**. The control system **26** is configured to manage the temperature of the heat load **16** so that the heat load **16** does not exceed threshold temperatures, i.e. a predetermined heat load temperature. In some embodiments, the heat load **16** may comprise of multiple heat loads **16** that may be arranged in series, parallel, or a combination. The control system **26** is also configured to manage the flow of fluid in the cooling loop **20** so that cooling of the heat load **16** is provided in the most efficient way. In other words, the control system **26** is configured to manage the flow of fluid so as to minimize the fuel burned to complete the flight cycle. The thermal-storage fuel system **24** adds the capability to reduce the amount of heat rejected to the environment and to determine the most efficient time to reject the heat to the environment.

(18) The control system **26** includes valves **52**, **53**, **54**, **55**, **57**, sensors **62**, **63**, **64**, **66**, **67**, and a

controller **68** as shown in FIGS. 2 and 3. The valves **52**, **53** are configured to transfer fuel between the fuel tanks **44**, **46** to cool the fuel in the first fuel tank **44** or heat the fuel in the second fuel tank **46**. The valve **54** is connected to the fluid conduit **30** downstream of the air-stream heat exchanger **36** and configured to selectively cause a flow of fluid to bypass the fuel-tank heat exchanger **48**. The valves **55**, **57** are configured to vary the flow rate of the fluid in the cooling loop **20** through the air-stream heat exchanger **36**. The sensor **62** is configured to measure the temperature of the fuel at the inlet of the fuel pump **47**, while the sensor **63** is configured to measure the temperature of the fuel at the outlet of the fuel pump **47**. The sensor **64** is configured to measure the temperature of the fluid in the fuel tank **46**. The sensor **66** is configured to measure the temperature of the fluid in the cooling loop **20** directly upstream of the fuel-tank heat exchanger **48** as it enters the fuel-tank heat exchanger **48**. The sensor **67** is configured to measure the temperature of the fluid in the cooling loop **20** as it returns to an accumulator **38** included in the cooling loop **20**. In some embodiments, where the cooling loop **20** is a two-phase flow system, the sensor **67** may be a pressure gauge.

(19) The controller **68** is configured to manage the cooling of the heat loads **16** as well as the use of the thermal-storage fuel system **24**. The controller **68** is configured to direct the valves **54**, **55**, **57** to vary the flow of the fluid in the cooling loop **20**. The controller **68** is also configured to control the fuel pump flow rate by the fuel pump **47** included in the thermal-storage fuel system **24** through the fuel-tank heat exchanger **48**. The controller **68** controls the valves **54**, **55**, **57** and the fuel pump **47** so as to maintain the fluid in the cooling loop **20** at a predetermined temperature.

(20) The controller **68** is configured to maximize the fuel savings of the aircraft **10** based on several factors, such as the type of day (hot or cold), the temperature along the flight path. In the illustrative embodiment, the controller **68** is configured to gather the state of the thermal energy system **12** prior to the flight as well as expected conditions along the flight path. Before take-off, the controller **68** is configured to assess the amount of fuel on board the aircraft **10** in each of the first fuel tank **44** and the second fuel tank **46**, as well as temperature of each tank **44**, **46**. This establishes the initial heat sink capacity on board the aircraft **10**.

(21) Additionally, the controller **68** is configured to analyze the anticipated flight path, including speed, altitude, air conditions (temperature, wind speed) to determine the anticipated change in weight of the fuel over the flight profile and the expected temperature of the air heat sink. The controller **68** is configured to use flight path conditions to determine when heat should be added to the first fuel tank **44** and when it would be best to remove heat from the first fuel tank **44** to minimize fuel burn.

(22) The control system **26** may be configured with model-based controls that run simulations prior to the flight. These simulations may be achieved through the use of artificial intelligence, general rules, or physics based calculations. The control system **26** may be configured to generate a schedule of heat to be rejected to the first fuel tank **44** as a function of location in the flight path.

(23) In some embodiments, the controller **68** is configured to transfer as much heat into the fuel as possible, while not exceeding any system thermal limits. In this way, a greater percentage of the waste heat to be removed may be returned to the engine cycle to produce useful work. Therefore, on a cold day, the amount of heat transferred to the first fuel tank **44** may be much larger than on a hot day.

(24) In addition, the controller **68** may be configured to minimize drag forces on the aircraft **10**. With a variable air stream **35** passing through the air-stream heat exchanger **36**, the cooling in the air-stream heat exchanger **36** may be timed to minimize fuel burn over the mission.

(25) Using this information, a pre-flight fuel tank heat rejection schedule may be generated to control the flow of fluid in the cooling loop **20** to the thermal-storage fuel system **24**. The schedule also includes re-cooling of the fuel tanks **44**, **46** in preparation for being able to burn the fuel stored in the first fuel tank **44** and to also provide cooling during descent and taxiing after landing. This could require significant cooling of the fuel tanks **44**, **46** while still at altitude.

(26) In addition, the control system **26** may be configured to mix fuel between the first and second fuel tanks **44**, **46** to maximize the use of the fuel thermal storage. Additionally, the pre-flight schedules may be updated during the flight.

(27) Once the pre-flight schedules are established, the aircraft **10** may take-off. As the flight proceeds, the temperature of the fuel in the first fuel tank **44** is measured by the sensors **62**, **63**. The upstream sensor **62** measures the temperature of the fuel flowing through the fuel pump **47** and the downstream sensor **63** measures the temperature of the fuel exiting the fuel-tank heat exchanger **48**. The controller **68** is configured to determine the temperature rise on the fuel stream going through the fuel-tank heat exchanger **48** based on the temperature measurements by the sensors **62**, **63**. The controller **68** is also configured to measure or calculate the fuel flow rate based on operating conditions of the fuel pump **47**.

(28) In this way, the controller **68** measures the actual heat rejection or heat absorption between the fluid and the fuel in the first fuel tank **44**. The control system **26** is configured to match the actual heat transfer between the fluid and the fuel in the first fuel tank **44** with the heat rejection schedule.

(29) To match the actual heat transfer between the fluid and the fuel in the first fuel tank **44**, the controller **68** may direct the valves **55**, **57** to vary the flow of fluid through the air-stream heat exchanger **36**. The controller **68** may also direct the valve **54** to vary the flow of fluid through the fuel-tank heat exchanger **48**. Alternatively, the controller **68** may be configured to vary the fuel flow rate through the fuel-tank heat exchanger **48** by changing fuel pump speed of the fuel pump **47**.

(30) In some embodiments, the full heat load **16** may not be able to be rejected to the fuel in the first fuel tank **44**. In that case, the remaining heat load **16** may be rejected through the air-stream heat exchanger **36**.

(31) The amount of heat to be rejected through the air-stream heat exchanger **36** may be set to satisfy the temperature requirement measured by sensors **66**, **67**. The amount of heat to be rejected through the air-stream heat exchanger **36** may also be driven by the temperature of the fuel in the first fuel tank **44**. If the fuel in the first fuel tank **44** is approaching thermal limits of the first fuel tank **44**, the flow of cooling fluid to the first fuel tank **44** may be used to cool the fuel in the first tank **44**.

(32) In some embodiments, the temperature of the fuel in the first fuel tank **44** may drive the exit temperature from the air-stream heat exchanger **36** measured by the sensor **66**. In such embodiments in which heat is to be absorbed from the fuel in the first fuel tank **44**, the temperature measured by the sensor **66** should be lower than the temperature in the first fuel tank **44** and may be controlled to a target temperature to achieve the desired heat absorption from the first fuel tank **44**.

(33) In some embodiments, the controller **68** is configured to vary the amount of air flowing through the air-stream heat exchanger **36** in the duct **34**. Reducing the airflow rate may also reduce the drag on the aircraft **10**.

(34) In some embodiments, the controller **68** is configured to direct the modulating valves **55**, **57** to vary the coolant flow rate through the air-stream heat exchanger **36**. In other embodiments, one of the modulating valves **55** may be replaced with a fixed flow restriction. The sensor **66** and/or the sensor **67** is configured to measure the temperature of the fluid to enable the controller **68** to better manage the heat rejection from the air-stream heat exchanger **36**.

(35) In some embodiments, the controller **68** is configured to control the flow of the fluid in the cooling loop **20** through the fuel-tank heat exchanger **48** based on a temperature measured by the sensor **62**. The controller **68** may direct the valve **54** to allow the fluid in the cooling loop **20** through the fuel-tank heat exchanger **48** in response to the temperature of the fuel in the first fuel tank **44** being below a first fuel tank predetermined threshold temperature. The controller **68** may direct the valve **54** to fully, or partially block the fluid in the cooling loop **20** through the fuel-tank heat exchanger **48** in response to the temperature of the fuel in the first fuel tank **44** being above the

first fuel tank predetermined threshold temperature.

(36) In some embodiments, the first fuel tank predetermined threshold temperature may be the maximum allowable tank temperature of the first fuel tank **44**. In some embodiments, the first fuel tank predetermined threshold temperature may be the maximum allowable fuel temperature at which the fuel can be safely supplied to the engine **14**. In some embodiments, the first fuel tank predetermined threshold temperature may be changed throughout the flight mission.

(37) In typical fuel systems, as fuel is passed to the engine **14**, the aircraft **10** may add heat to the fuel. The engine **14** may further heat the fuel before it is burned in the combustor. Often, the engine **14** may have a maximum allowable fuel temperature at which the fuel may be supplied to the engine **14** that is far lower than the maximum temperature that the fuel will be when it is burned. This temperature limit may be used to ensure the fuel entering the engine **14** is cold enough to cool engine heat loads before burned in the combustor.

(38) In some embodiments, the first fuel tank **44** may be configured to allow the fuel in the first fuel tank **44** to be heated to the maximum temperature of the fluid in the cooling loop **20**. In other embodiments, the first fuel tank **44** may be configured to allow the fuel in the first fuel tank **44** to be heated to some other temperature limit that is above the maximum allowable fuel temperature that may be delivered to the engine. The second fuel tank **46** stores fuel for the engine **14** at a temperature equal to or less than the maximum allowable fuel temperature that may be delivered to the engine **14**. The second fuel tank is therefore configured to provide the fuel at the maximum allowable fuel temperature of the engine **14**.

(39) If the temperature of the fuel in the first fuel tank **44** is approaching the first fuel tank predetermined threshold temperature, one of the valves **52**, **53** may allow some fuel to transfer between the fuel tanks **44**, **46** to cool the fuel in the first fuel tank **44** or heat the fuel in the second fuel tank **46**. As such, the controller **68** may determine if the valve **52** may transfer fuel between the fuel tanks **44**, **46** based at least partially on the temperature measured by the sensors **62**, **63**, **64**.

(40) The valves **52**, **53** provide fluid communication between the first and second fuel tanks **44**, **46**. In some embodiments, the first fuel tank **44** may have an inlet valve **53** and outlet valve **52** that are fluidly connected to the second fuel tank **46**. The hot fuel in the first fuel tank **44** exits the first fuel tank **44** through the outlet **52** to the second fuel tank **46**, while the relatively cooler fuel enters the first fuel tank **44** through the inlet valve **53** from the second fuel tank **46**. The exchanger of the fuel between the first and second fuel tanks **44**, **46** may be assisted with the fuel pump **47** through various piping arrangements and valves.

(41) The thermal-storage fuel system **24** may further include another pump in fluid communication between the tanks **44**, **46**. The second pump may be configured to move the fuel between the tanks **44**, **46**.

(42) In other embodiments, the system **12** may include a single fuel tank. The single fuel tank may be coupled to engine **14** to deliver the fuel to the engine **14**. In such embodiments, the first fuel tank predetermined threshold temperature is a single fuel tank predetermined threshold temperature. In such embodiments, the single fuel tank predetermined threshold temperature is the maximum allowable fuel temperature at which the fuel can be safely passed to the engine **14**.

(43) In some embodiments, the controller **68** may also be configured to control the flow of the fluid in the cooling loop **20** through the fuel-tank heat exchanger **48** based on the temperature measured by the sensors **62**, **63**, **66**, **67**. The controller **68** may be configured to direct the valve **54** to control the fluid in the cooling loop **20** through the fuel-tank heat exchanger **48** in response to a temperature measured by at least one of the sensors **62**, **63**, **66**, **67**.

(44) If the temperature of the fluid directly upstream of the fuel-tank heat exchanger **48** measured by the sensor **66** is below the first fuel tank predetermined threshold temperature, while the temperature of the fuel in the first fuel tank **44** is equal to or exceeds the first fuel tank predetermined threshold temperature, the controller **68** may allow the flow of fluid to the fuel-tank heat exchanger **48**. Conversely, if the temperature of the fluid directly upstream of the fuel-tank

heat exchanger **48** is above the first fuel tank predetermined threshold temperature, while the temperature of the fuel in the first fuel tank **44** is equal to or exceeds the first fuel tank predetermined threshold temperature, the controller **68** may fully, or partially block the flow of fluid to the fuel-tank heat exchanger **48**.

(45) In other embodiments, if the temperature of the fluid returning to the accumulator **38** measured by the sensor **67** is below a sensor target temperature, and the first fuel tank temperature is greater than the sensor target temperature, the controller **68** may direct the valve **54** to allow the flow of fluid to the fuel-tank heat exchanger **48**. Conversely, if the temperature measured by the sensor **67** is above the sensor target temperature, the controller **68** may direct the valve **54** to fully, or partially block the flow of fluid to the fuel-tank heat exchanger **48**.

(46) During use of the thermal energy system **12** in the aircraft **10**, the fluid in the fluid conduit **30** is pumped from the accumulator **38** to the heat source **16** as indicted by arrow **80**. The pump **32** moves the fluid through the fluid conduit **30** to transfer heat from the heat source **16** to the fluid to cool the heat source **16** as indicted by arrow **82**. Moving from the heat source **16**, the flow of fluid in the fluid conduit **30** moves through the air-stream heat exchanger **36** to transfer heat between the fluid in the cooling loop **20** and the air passing through the duct **34** of the cooler **22** as indicated by arrow **84**. The modulating valves **55**, **57** are configured to vary the flow of fluid through the air-stream heat exchanger **36** based at least in part on the temperature measured by the sensor **62**, **63**, **64**, **66**, **67**.

(47) As the fluid in the fluid conduit **30** exits the air-stream heat exchanger **36** as indicated by arrow **86**, the valve **54** controls the flow of fluid to the thermal-storage fuel system **24**. The valve **54** is connected to the fluid conduit **30** downstream of the air-stream heat exchanger **36** to selectively cause the flow of fluid to fully, or partially bypass the fuel-tank heat exchanger **48** based at least in part on the temperature measured by the sensors **62**, **63**, **64**, **66**, **67**.

(48) The controller **68** may be configured to direct the valve **54** to allow the flow of the fluid in the fluid conduit **30** through the fuel-tank heat exchanger **48** based on the temperature measured by one or all of the sensors **62**, **63**, **64** as shown in FIG. 2. The valve **54** allows the fluid in the fluid conduit **30** to flow to the fuel-tank heat exchanger **48** as indicated by arrow **88** in FIG. 2. The valve **54** allows the fluid to flow through the fuel-tank heat exchanger **48** in response to the temperature of the fuel in the first fuel tank **44** being below the first fuel tank predetermined threshold temperature.

(49) Additionally, the valve **54** may also allow the fluid in the fluid conduit **30** to flow to the fuel-tank heat exchanger **48** in response to the temperature of the fluid in the cooling loop **20** directly upstream of the fuel-tank heat exchanger **48**. If the temperature of the fluid directly upstream of the fuel-tank heat exchanger **48** is below the first fuel tank predetermined threshold temperature, while the temperature of the fuel in the first fuel tank **44** is equal to or exceeds the first fuel tank predetermined threshold temperature, the valve **54** allows the flow of fluid to the fuel-tank heat exchanger **48**. As the fluid in the fluid conduit **30** exits the fuel-tank heat exchanger **48**, the flow of fluid returns to the accumulator **38** as indicated by arrow **90** in FIG. 2.

(50) During taxiing or take-off of the aircraft **10**, the air flowing through the ram air cooler **22** may be about the same temperature of the fluid in the fluid conduit **30**. As such, the heat transfer between the fluid and the air-stream heat exchanger **36** is low such that the heat of the fluid in the cooling loop **20** is not sufficiently removed. To expel heat from the system **12**, the valve **54** may allow the fluid in the fluid conduit **30** to flow to the fuel-tank heat exchanger **48** to transfer heat from the fluid in the cooling loop **20** to the fuel in the first fuel tank **44** as shown in FIG. 2.

(51) In the illustrative embodiment, as the valve **54** allows fluid in the fluid conduit **30** to flow to the fuel-tank heat exchanger **48**, the valve **54** may fully, or partially, block flow from bypassing the fuel-tank heat exchanger **48** as suggested by the dotted line **92** in FIG. 2. As such, little to no fluid is flowing in the fluid conduit **30** in the region of the dotted line **92** as shown in FIG. 2.

(52) In other embodiments, the valve **54** may partially allow fluid in the fluid conduit **30** to flow

through the fuel-tank heat exchanger **48**. As such, some fluid may be flowing to the fuel-tank heat exchanger **48** and to the accumulator **38**. In such embodiments, the lines **88**, **90**, **92** would be solid to show the flow of fluid in fluid conduit **30** to each of the components **48**, **38**.

(53) In the illustrative embodiment, the ram air cooler **22** may be bypassed during taxiing or take off to reduce the amount of drag created by the duct **34**. As the majority, if not all, of the heat from the heat source **16** is transferred through the fuel-tank heat exchanger **48**; the ram air cooler **22** may be closed off to block air flowing through the ram air cooler **22**, to minimize drag.

(54) In the illustrative embodiment, the ram air cooler **22** further includes a variable NACA inlet **33** as shown in FIGS. **2** and **3**. The variable NACA inlet **33** will reduce the amount of drag created from the duct **34** when the ram air cooler **22** is not being used. The variable inlet **33** varies the amount of air from the atmosphere **18** through the air-stream heat exchanger **36** as indicated by arrow **35**.

(55) As the fluid in the fluid conduit **30** heats the fuel in the first fuel tank **44**, the temperature of the fuel approaches the first fuel tank predetermined threshold temperature. However, at higher altitudes, the air is cooler in temperature and therefore the air flowing through the ram air cooler **22** is viable for transferring heat from the fluid to the air. Therefore, as the aircraft **10** increases in altitude and/or speed, the air-stream heat exchanger **36** may begin to transfer more heat from the fluid in the cooling loop **20** to the atmosphere air flowing through the duct **34**.

(56) To manage the temperature of the fuel in the first fuel tank **44** and the fluid in the fluid conduit **30**, the controller **68** directs the valve **54** to fully, or partially block the flow of the fluid through the fuel-tank heat exchanger **48** based on the temperature measured by the sensors **62**, **63**, **64**, **66**, **67** as shown in FIG. **3**. If the temperature of the fluid measured by the sensor **66** is above the first fuel tank predetermined threshold temperature, while the temperature of the fuel in the first fuel tank **44** is equal to or exceeds the first fuel tank predetermined threshold temperature, the controller **68** may fully, or partially block the flow of fluid to the fuel-tank heat exchanger **48**. Dotted arrows **88**, **90**, indicate the blocked flow of fluid to the fuel-tank heat exchanger **48**.

(57) If the valve **54** blocks the fluid in the cooling loop **20** from flowing to the fuel-tank heat exchanger **48**, the fluid flows back to the accumulator **38** as suggested by the arrow **92** in FIG. **3**. However, as the air-stream heat exchanger **36** begins to transfer more heat to the air, the fluid in the cooling loop **20** may be cool enough to transfer, or absorb heat from the fuel in the first fuel tank **44**.

(58) As such, the valve **54** allows the flow of the fluid in the fluid conduit **30** through the fuel-tank heat exchanger **48** if the temperature of the fluid measured by the sensor **66** is below the first fuel tank predetermined threshold temperature, while the temperature of the fuel in the first fuel tank **44** is equal to or exceeds the first fuel tank predetermined threshold temperature. In this way, the temperature of the fuel in the first fuel tank **44** is reduced in preparation to act as a heat sink later in the flight, as the aircraft **10** descends into warmer air. In such environments, the air-stream heat exchanger **36** may not be able to reject as much heat to the air.

(59) A method of cooling heat loads on an aircraft **10** may include several steps. The method includes conducting a cooling fluid through the cooling loop **20** to transfer heat between the heat source **16** and the fluid in the cooling loop **20**, conducting the fluid in the cooling loop **20** to the air-stream heat exchanger **36** to transfer heat between the fluid in the cooling loop **20** and the air passing through the duct **34**, and conducting the fluid in the cooling loop **20** to the fuel-tank heat exchanger **48** to transfer heat between the fluid in the cooling loop **20** and fuel stored in the first fuel tank **44**. In this way, the fuel in the first and second fuel tanks **44**, **46** is not transported to the heat loads **16**, but rather, the fluid is brought to the fuel to transfer the heat with the fuel.

Transferring heat with the fluid to the fuel in the first fuel tank **44** minimizes safety risks, and heating the fuel prior to burning in the engine **14** may provide additional benefits.

(60) In the illustrative embodiment, the method further includes selectively varying a flow of the fluid in the cooling loop **20** through the air-stream heat exchanger **36** and a flow of the fluid

through the fuel-tank heat exchanger **48**. Varying the flows of the fluid through the air-stream heat exchanger **36** and the fuel-tank heat exchanger **48** maintains a temperature of the heat source **16** below a predetermined heat load temperature.

(61) In the illustrative embodiment, the method further includes measuring the heat transfer between the fluid and the fuel in the first fuel tank **44** and comparing the heat transfer measured to a predetermined heat rejection schedule. Based on the comparison, the flow of fluid in the cooling loop **20** through the fuel-tank heat exchanger **48** is selectively varied in response to the heat transfer measured being different from the predetermined heat rejection schedule.

(62) In the illustrative embodiment, the method further includes measuring the temperature of the fluid in the cooling loop **20** upstream of the accumulator **38** included in the cooling loop **20** using the sensor **67**. The temperature measured is then compared to the sensor target temperature and the flow of fluid in the cooling loop **20** is selectively varied through the air-stream heat exchanger **36** and fuel tank heat exchanger **48** in response to the temperature measured being different from the sensor target temperature.

(63) Another embodiment of a thermal energy system **212** in accordance with the present disclosure is shown in FIG. **4**. The thermal energy system **212** is substantially similar to the thermal energy system **12** shown in FIGS. **1-3** and described herein. Accordingly, similar reference numbers in the **200** series indicate features that are common between the thermal energy system **212** and the thermal energy system **12**. The description of the thermal energy system **12** is incorporated by reference to apply to the thermal energy system **212**, except in instances when it conflicts with the specific description and the drawings of the thermal energy system **212**.

(64) The thermal energy system **212** includes a cooling loop **220**, a cooler **222**, a thermal-storage fuel system **224**, and a control system **226** as shown in FIG. **4**. The cooling loop **220** has a fluid conduit **230** and a pump **232** configured to move fluid through the fluid conduit **230** to transfer heat from the heat source **216** to the fluid thereby cooling the heat source **216**. The cooler **222** includes a duct **234** configured to conduct air, arrow **235**, therethrough and an air-stream heat exchanger **236** located in the duct **234**. The air-stream heat exchanger **236** is in thermal communication with the fluid conduit **230** to transfer heat between the fluid in the cooling loop **220** and the air passing through the duct **234**. The thermal-storage fuel system **224** is configured to transfer heat between the fluid in the cooling loop **220** and fuel stored in the thermal-storage fuel system **224**. The control system **226** is configured to control the flow of fluid through the cooling loop **220** to manage the cooling of the heat loads **216** as well as manage the use of the thermal-storage fuel system **224**.

(65) The thermal-storage fuel system **224** includes fuel tanks **244**, **246**, a fuel-tank heat exchanger **248**, and an engine-fuel unit **250** as shown in FIG. **4**. Both the first fuel tank **244** and second fuel tank **246** are configured to store fuel therein, while the second fuel tank **246** is in fluid communication with the first fuel tank **244** through the valves **252**, **253** to transfer fuel selectively therebetween. The fuel-tank heat exchanger **248** is in thermal communication with the fluid conduit **230** and is configured to transfer heat between the fluid in the cooling loop **220** and fuel stored in the first fuel tank **244**. The engine-fuel unit **250** is configured to receive fuel from the second fuel tank **246** and deliver the fuel to the engine **214**. In the illustrative embodiment, the engine-fuel unit **250** is in thermal communication with the cooling loop **220** to transfer heat between the fluid in the cooling loop **220** and the fuel in the engine-fuel unit **250**.

(66) The engine-fuel unit **250** includes a mix valve **274**, conduits **276**, **278**, and an engine-fuel heat exchanger **279** as shown in FIG. **4**. The first conduit **276** is in fluid communication between the mix valve **274** and the second fuel tank **246** to deliver fuel having a first temperature to the mix valve **274**. The second conduit **278** is in fluid communication between the mix valve **274** and the second fuel tank **246**. The engine-fuel heat exchanger **279** is in thermal communication with the second conduit **278** to cause the second conduit **278** to deliver fuel having a second temperature to the mix valve **274**.

(67) In the illustrative embodiment, the mix valve **274** and the conduits **276**, **278** form a valve

system in fluid communication with the second fuel tank **246** and the engine-fuel heat exchanger **279**. The valve system, including the mix valve **274** and the conduits **276**, **278**, is configured to vary a flow of fuel through the engine-fuel heat exchanger **279** to deliver fuel to the engine **214** at an engine-fuel unit predetermined threshold temperature.

(68) In the illustrative embodiment, the engine-fuel unit predetermined threshold temperature is the desired temperature for the fuel delivered to the engine **214** from the valve system. In some embodiments, the engine-fuel unit predetermined threshold temperature may be equal to a maximum allowable fuel temperature. The maximum allowable fuel temperature is the maximum temperature that the fuel may be when it is passed to the engine **214**.

(69) In some embodiments, the mix valve **274** is configured to control the flow of fuel to the engine-fuel heat exchanger **279** to match the maximum allowable fuel temperature. The maximum allowable fuel temperature may vary throughout the flight profile or be a single value.

(70) In some embodiments, the mix valve **274** is configured to vary a first flow rate of fuel from the first conduit **276** and a second flow rate of fuel from the second conduit **278**. The mix valve **274** varies the first and second flow rates for the respective conduits **276**, **278** to provide a mixed stream of fuel having a third temperature. The third temperature may also be equal to or less than the maximum allowable fuel temperature, which is the temperature at which the fuel can be safely supplied to the engine **214**.

(71) The maximum allowable tank temperature of the second fuel tank **246** may be less than the maximum allowable fuel temperature. In this way, the fuel in the second fuel tank **246** may absorb heat from the heat loads **216** and be below the maximum allowable fuel temperature.

(72) In the illustrative embodiments, the maximum allowable tank temperature of the second fuel tank **246** may be less than the maximum allowable tank temperature of the first fuel tank **244**. The maximum allowable fuel temperature, i.e. the temperature at which the fuel can be safely supplied to the engine **14**, may be equal to or greater than the maximum allowable tank temperature of the second tank **246**.

(73) The maximum allowable tank temperature of the first fuel tank **244** may be greater than or equal to the maximum allowable fuel temperature. The fuel in the first fuel tank **244** may be heated to a temperature greater than that of the maximum allowable fuel temperature for the engine **214** because the first fuel tank **244** may be cooled during the mission before the fuel in the first fuel tank **244** may need to be sent to the engine **214**.

(74) In some embodiments, the engine-fuel unit predetermined threshold temperature may be equal to or greater than the first fuel tank predetermined threshold temperature. In some embodiments, the engine-fuel unit predetermined threshold temperature may vary throughout the flight mission.

(75) Turning again to the control system **226**, the control system **226** includes the valves **254**, **255**, **257**, sensors **262**, **263**, **264**, **266**, **267**, and a controller **268** as shown in FIG. **4**. The valve **254** is connected to the fluid conduit **230** and configured to selectively cause a flow of fluid to bypass the fuel-tank heat exchanger **248**. The valves **255**, **257** are connected to the fluid conduit **230** and configured to selectively vary the flow through the heat exchanger **236**. The sensor **262** is configured to measure the temperature of the fuel at the inlet of a fuel pump **247**, while the sensor **263** is configured to measure the temperature of the fuel at the outlet of the fuel pump **47**. The sensor **264** is configured to measure the temperature of the fluid in the fuel tank **246**. The sensor **266** is configured to measure the temperature of the fluid in the cooling loop **20** directly upstream of the fuel-tank heat exchanger **248** as it enters the fuel-tank heat exchanger **248**. The sensor **267** is configured to measure the temperature of the fluid in the cooling loop **220** as it returns to an accumulator **238** included in the cooling loop **220**.

(76) The controller **268** is configured to manage the cooling of the heat loads **216** as well as the use of the thermal-storage fuel system **224**. The controller **268** is configured to direct the valves **254**, **255**, **257** to vary the flow of the fluid in the cooling loop **220**. The controller **268** is also configured to control the fuel pump flow rate by the fuel pump **247** included in the thermal-storage fuel system

224 through the fuel-tank heat exchanger 248. The controller 268 controls the valves 254, 255, 257 and the fuel pump 247 so as to maintain the fluid in the cooling loop 220 at a predetermined temperature.

(77) In the illustrative embodiments, the controller 268 is configured to allow the flow of the fluid in the cooling loop 220 through the fuel-tank heat exchanger 248 based on a temperature measured by the sensor 262. The controller 268 directs the valve 254 to allow the fluid in the cooling loop 220 through the fuel-tank heat exchanger 248 in response to a temperature of the fuel in the first fuel tank 244 being below the first fuel tank predetermined threshold temperature.

(78) In the illustrative embodiments, the controller 268 is also configured to allow the flow of the fluid in the cooling loop 220 through the fuel-tank heat exchanger 248 based on the temperature measured by the sensors 262, 263, 266, 267. The controller 268 may be configured to direct the valve 254 to control the fluid in the cooling loop 220 through the fuel-tank heat exchanger 248 in response to a temperature measured by at least one of the sensors 262, 263, 266, 267.

(79) If the temperature of the fluid directly upstream of the fuel-tank heat exchanger 248 is below the first fuel tank predetermined threshold temperature, while the temperature of the fuel in the first fuel tank 244 is equal to or exceeds the first fuel tank predetermined threshold temperature, the controller 268 allows the flow of fluid to the fuel-tank heat exchanger 248. Conversely, if the temperature of the fluid directly upstream of the fuel-tank heat exchanger 248 is above the first fuel tank predetermined threshold temperature, while the temperature of the fuel in the first fuel tank 244 is equal to or exceeds the first fuel tank predetermined threshold temperature, the controller 268 may fully, or partially block the flow of fluid to the fuel-tank heat exchanger 248.

(80) In other embodiments, if the temperature of the fluid returning to the accumulator 238 measured by the sensor 267 is below the sensor target temperature, the controller 268 may direct the valve 254 to allow the flow of fluid to the fuel-tank heat exchanger 248, if the fuel tank temperature is higher than the sensor target temperature. Conversely, if the temperature measured by the sensor 267 is above the sensor target temperature, the controller 268 may direct the valve 254 to fully, or partially block the flow of fluid to the fuel-tank heat exchanger 248 if the fuel tank temperature is higher than the sensor temperature. However, if the fuel tank temperature is lower than the sensor target temperature, valve 254 will pass flow to the fuel tank to provide cooling to the fluid.

(81) During use of the thermal energy system 212 in the aircraft 10, the fluid in the fluid conduit 230 is pumped from the accumulator 238 included in the cooling loop 220 to the heat source 216 as indicted by arrow 280. The pump 232 moves the fluid through the fluid conduit 230 to transfer heat from the heat source 216 to the fluid to cool the heat source 216 as indicted by arrow 282. Moving from the heat source 216, the flow of fluid in the fluid conduit 230 moves through the engine-fuel heat exchanger 279 of the engine-fuel unit 250 as indicated by arrow 283. The fluid in the fluid conduit 230 is in fluid communication with the engine-fuel heat exchanger 279 to transfer heat between the fluid and the fuel flowing to the engine 214.

(82) As the fluid in the fluid conduit 230 exits the engine-fuel unit 250, the fluid flows to the ram-air heat exchanger 236 as indicated by arrow 284. The fluid flows through the ram-air heat exchanger 236 to transfer heat between the fluid in the cooling loop 20 and the air passing through the duct 34 of the ram air cooler 22.

(83) As the fluid in the fluid conduit 230 exits the ram-air heat exchanger 236 as indicated by arrow 286, the valve 254 controls the flow of fluid to the thermal-storage fuel system 224. The valve 254 is configured to selectively cause the flow of fluid to bypass the fuel-tank heat exchanger 248 based at least in part on the temperature measured by the sensors 262, 263, 264, 266, 267.

(84) The valve 254 may allow the fluid in the fluid conduit 230 to flow to the fuel-tank heat exchanger 248 in response to the temperature of the fuel in the first fuel tank 244 being below the first fuel tank predetermined threshold temperature as indicated by arrow 288. The valve 254 may also allow the fluid to flow to the fuel-tank heat exchanger 248 in response to the temperature of

the fluid directly upstream of the fuel-tank heat exchanger **248** being below the first fuel tank predetermined threshold temperature, while the temperature of the fuel in the first fuel tank **44** is equal to or exceeds the first fuel tank predetermined threshold temperature.

(85) In the illustrative embodiment, as the valve **254** allows fluid in the fluid conduit **230** to flow to the fuel-tank heat exchanger **248**, the valve **254** blocks flow from bypassing the fuel-tank heat exchanger **248** as suggested by the dotted line **292** in FIG. **4**. As such, no fluid is flowing in the fluid conduit **230** in the region of the dotted line **292**. As the fluid in the fluid conduit **230** exits the fuel-tank heat exchanger **248**, the flow of fluid returns to the accumulator **238** as show by arrow **290**.

(86) As discussed above, the controller **268** is configured to direct the valve **254** to fully, or partially block the flow of the fluid in the fluid conduit **230** through the fuel-tank heat exchanger **248** based on the temperature measured by the sensors **262**, **263**, **264**, **266**, **267**. If the valve **254** partially or fully blocks the fluid in the fluid conduit **230** from flowing to the fuel-tank heat exchanger **248**, the rest or all of fluid flows to the accumulator **238**. As such, little to no fluid is flowing in the fluid conduit **230** in and out of the fuel-tank heat exchanger **248**.

(87) As the fluid in the fluid conduit **230** heats the fuel in the first fuel tank **244**, the temperature of the fuel approaches the first fuel tank predetermined threshold temperature. As such, the fuel in the second fuel tank **246** may be used as an additional heat sink to prevent the temperature of the fuel in the first fuel tank **244** from exceeding the first fuel tank predetermined threshold temperature.

(88) The fluid in the fluid conduit **230** flowing to the engine-fuel heat exchanger **279** heats the fuel in the second conduit **278** to cool the fluid in the fluid conduit **230**. The mix valve **274** varies the first flow rate of fuel from the first conduit **276** and the second flow rate of fuel from the second conduit **278** to provide the mixed stream of fuel with the third temperature. In the illustrative embodiments, the controller **268** control the flow of fluid to put as much heat into the fuel flowing through the engine-fuel heat exchanger **279** without exceeding the engine-fuel unit predetermined threshold temperature.

(89) However, at higher altitudes, the air is cooler in temperature and therefore the air flowing through the cooler **222** is viable for transferring heat between the fluid and the air. Therefore, as the aircraft **10** increases in altitude, the ram-air heat exchanger **236** may begin to transfer more heat between the fluid in the cooling loop **220** and the atmosphere air flowing through the duct **234**.

(90) As the engine-fuel unit **250** and the ram-air heat exchanger **236** begin to transfer more heat out of the system, the fluid in the cooling loop **220** may be cool enough to transfer heat from the fuel in the first fuel tank **244**. Thus, during flight, the temperature of the first fuel tank **244** is lowered below the maximum allowable tank temperature of the first tank **244**. This cooling process allows the first fuel tank **244** to be able to absorb more heat from the fluid in the cooling loop **220** later in the flight cycle when cooling the fluid in the fluid conduit **230** may be more challenging. In this way, the temperature of the fuel in the first fuel tank **244** is reduced in preparation to act as a heat sink later in the flight, as the aircraft **10** descends into warmer air. In such environments, the air-stream heat exchanger **236** may not be able to reject as much heat to the air.

(91) In the illustrative embodiment, the fluid in the cooling loop is conducted to the engine-fuel unit before it is conducted to the air-stream heat exchanger. The fluid in the cooling loop is conducted to the engine-fuel unit before the air-stream heat exchanger to transfer heat between the fluid in the cooling loop and the fuel in the engine-fuel unit.

(92) A method of cooling heat loads on the aircraft **10** may include several steps and may be substantially similar to the method described in regards to the embodiment shown in FIGS. **1-3**. The method further includes varying the flow of fuel through the engine-fuel heat exchanger **279** to deliver fuel to the engine **214** at the engine-fuel unit predetermined threshold temperature. The mix valve **274** varies the first flow rate of fuel from the first conduit **276** and the second flow rate of fuel from the second conduit **278** to provide the mixed stream of fuel. The mixed stream of fuel has the third temperature that is less than or equal to the engine-fuel unit predetermined threshold

temperature.

(93) Another embodiment of a thermal energy system **312** in accordance with the present disclosure is shown in FIG. 5. The thermal energy system **312** is substantially similar to the thermal energy system **12** shown in FIGS. 1-3 and described herein. Accordingly, similar reference numbers in the **300** series indicate features that are common between the thermal energy system **312** and the thermal energy system **12**. The description of the thermal energy system **12** is incorporated by reference to apply to the thermal energy system **312**, except in instances when it conflicts with the specific description and the drawings of the thermal energy system **312**.

(94) The thermal energy system **312** includes a cooling loop **320**, a cooler **322**, and a thermal-storage fuel system **324** as shown in FIG. 5. The cooling loop **320** has a fluid conduit **330** and a pump **332** configured to move fluid through the fluid conduit **330** to transfer heat from the heat source **316** to the fluid in the fluid conduit **330** thereby cooling the heat source **316**. The cooler **322** includes a duct **334** configured to receive air from the atmosphere **18** and an air-stream heat exchanger **336** located in the duct **334**. The air-stream heat exchanger **336** is in thermal communication with the fluid conduit **330** to transfer heat between the fluid in the cooling loop **320** and the air passing through the duct **334**. The thermal-storage fuel system **324** is configured to transfer heat between the fluid in the cooling loop **320** and fuel stored in the thermal-storage fuel system **324**.

(95) The fluid in the illustrative embodiment is oil, which is stored in an oil tank **338** included in the cooling loop **320** as shown in FIG. 5. In the other embodiments, the fluid of the cooling loop **20**, **220** is a refrigerant. In the illustrative embodiment of FIG. 5, the oil tank **338** replaces the accumulator using oil as the fluid in the cooling loop **320**.

(96) The thermal-storage fuel system **324** includes fuel tanks **344**, **346**, a fuel-tank heat exchanger **348**, and an engine-fuel unit **350** as shown in FIG. 5. Both the first fuel tank **344** and second fuel tank **346** are configured to store fuel therein, while the second fuel tank **346** is in fluid communication with the first fuel tank **344**. The fuel-tank heat exchanger **348** is in thermal communication with the fluid conduit **330** and is configured to transfer heat from the oil in the cooling loop **320** and fuel stored in the first fuel tank **344**. The engine-fuel unit **350** is configured to receive fuel from the second fuel tank **346** and deliver the fuel to the engine **314**. In the illustrative embodiment, the engine-fuel unit **350** is in thermal communication with the cooling loop **320** to transfer heat from the oil in the cooling loop **320** to the fuel in the engine-fuel unit **346**.

(97) Another embodiment of a thermal energy system **412** in accordance with the present disclosure is shown in FIGS. 6 and 7. The thermal energy system **412** is substantially similar to the thermal energy system **12** shown in FIGS. 1-3 and described herein. Accordingly, similar reference numbers in the **400** series indicate features that are common between the thermal energy system **412** and the thermal energy system **12**. The description of the thermal energy system **12** is incorporated by reference to apply to the thermal energy system **412**, except in instances when it conflicts with the specific description and the drawings of the thermal energy system **412**.

(98) The thermal energy system **412** includes a cooling loop **420**, a cooler **422**, a thermal-storage fuel system **424**, a control system **426**, a first heat source **416**, and a second heat source **428** as shown in FIGS. 6 and 7. The cooling loop **420** has a fluid conduit **430** and a pump **432** configured to move fluid through the fluid conduit **430** to transfer heat from the heat sources **416**, **428** to the fluid in the fluid conduit **430** thereby cooling the heat sources **416**, **428**. The cooler **422** includes a duct **434** configured to receive air from the atmosphere **18** and an air-stream heat exchanger **436** located in the duct **434** to transfer heat between the fluid in the cooling loop **420** and the air passing through the duct **434**. The thermal-storage fuel system **424** is configured to transfer heat between the fluid in the cooling loop **420** and fuel stored in the thermal-storage fuel system **424**. The control system **426** is configured to control the flow of fluid through the fluid conduit **430**.

(99) The first heat source **416** is a high temperature heat source **416**, while the second heat source **428** is a low temperature heat source **428**. The low temperature heat source **428** may need to be

kept cooler or at a lower temperature than the high temperature heat source **416**.

(100) In the illustrative embodiment, the low temperature device **428** is a battery **428**. In other embodiments, the low temperature device **428** may be another low temperature device, such as solid state electronics, a light-emitting diode (“LED”), an analog circuit, a digital circuit, a computer, a server, a server farm, a data center, a hoteling circuit such as vehicle electronics, a vehicle such as an aircraft, a directed-energy weapon, a laser, a plasma weapon, a microwave generator, a pulse-powered device, a satellite uplink, an electric motor or generator, an electric device, etc.

(101) In the illustrative embodiment, the cooling loop **420** is structured to conduct the fluid heated from the first heat source **416** and the second heat source **428** first through an engine-fuel heat exchanger **479**. Then the fluid in the cooling loop **420** is conducted through the air-stream heat exchanger **436** after the engine-fuel heat exchanger **479**. Then the fluid in the cooling loop **420** is conducted through a fuel-tank heat exchanger **448** after the air-stream heat exchanger **436**.

(102) In the illustrative embodiment, the cooling loop **420** further includes a plurality of expansion valves **440**, **441** and a compressor **442**. The first expansion valve **440** is arranged upstream of the low temperature device **428**. The second expansion valve **441** is arranged upstream of the thermal-storage fuel system **424**. The compressor **442** is arranged downstream of the low temperature device **428**.

(103) In the illustrative embodiment, the cooling fluid in the cooling loop **420** is a two-phase fluid. The expansion valves **440**, **441**, in conjunction with the compressor **442**, are configured to reduce the pressure of the fluid in the cooling loop **420** between the expansion valves **440**, **441** and the compressor **442**. By reducing the pressure of the fluid, the temperature is lowered, which allows the fluid to effectively absorb heat.

(104) In the illustrative embodiment, the cooling loop **420** is structured to conduct fluid through the second expansion valve **441** and through the fuel-tank heat exchanger **448** after the second expansion valve **441**. Then the cooling loop **420** conducts the fluid through the compressor **442** after the fuel-tank heat exchanger **448**, through the engine-fuel heat exchanger **479** after the compressor **442**, and through the air-stream heat exchanger **436** after the engine-fuel heat exchanger **479**.

(105) The thermal-storage fuel system **424** includes fuel tanks **444**, **446**, the fuel-tank heat exchanger **448**, and an engine-fuel unit **450** as shown in FIGS. **6** and **7**. Both the first fuel tank **444** and second fuel tank **446** are configured to store fuel therein, while the fuel-tank heat exchanger **448** is in thermal communication with the fluid conduit of the cooling loop **420** to transfer heat between the fluid in the cooling loop **420** and fuel stored in the first fuel tank **444**. The engine-fuel unit **450** is configured to receive fuel from the second fuel tank **446** and deliver the fuel to the engine **414**. In the illustrative embodiment, the engine-fuel unit **450** is in thermal communication with the cooling loop **420** to transfer heat between the fluid in the cooling loop **420** and the fuel in the engine-fuel unit **450**.

(106) The control system **426** includes a plurality of valves **454**, **455**, **456**, **457**, **458**, sensors **462**, **463**, **464**, **466**, **467**, and a controller **468** as shown in FIGS. **6** and **7**. The valves **454**, **455**, **456**, **457**, **458** are connected to the fluid conduit **430** and configured to selectively control the flow of fluid through the cooling loop **420**. The sensors **462**, **463**, **464**, **466**, **467** are configured to measure the temperature of the fuel in the fuel tanks **444**, **446** and the temperature of the fluid in the fluid conduit **430** of the cooling loop **420**. The controller **468** is configured to operate the plurality of valves **454**, **455**, **456**, **457**, **458** to allow the fluid to flow through the fluid conduit **430** based at least in part on the information measured by the sensors **462**, **463**, **464**, **466**, **467**.

(107) The controller **468** is configured to manage the cooling of the heat loads **416** as well as the use of the thermal-storage fuel system **424**. The controller **468** is configured to direct the valves **454**, **455**, **456**, **457**, **458** to vary the flow of the fluid in the cooling loop **420**. The controller **468** is also configured to control the fuel pump flow rate of the fuel pump **447** included in the thermal-

storage fuel system **424** through the fuel-tank heat exchanger **448**. The controller **468** controls the valves **454**, **455**, **456**, **457**, **458** and the fuel pump **447** so as to maintain the fluid in the cooling loop **420** at a predetermined temperature.

(108) The plurality of valves **454**, **455**, **456**, **457**, **458** includes a first valve **454**, a second valve **456**, a third valve **458**, and modulating valves **455**, **457** as shown in FIGS. **6** and **7**. The first valve **454** is connected to the fluid conduit **430** downstream of the air-stream heat exchanger **436** and is configured to selectively cause the flow of fluid to bypass the fuel-tank heat exchanger **448**. The second valve **456** is connected to the fluid conduit **430** downstream of the first valve **454** and the expansion valve **441** and is configured to selectively control the flow of fluid through the fuel-tank heat exchanger **448**. The third valve **458** is connected to the fluid conduit downstream of the fuel-tank heat exchanger **448** and is configured to selectively control the flow of fluid to the compressor **442** or back to the accumulator **438**. The valves **455**, **457** are configured to vary the flow rate of the fluid in the cooling loop **420** through the air-stream heat exchanger **436**.

(109) The control system **426** is configured to selectively vary a flow of the fluid in the cooling loop **420** through the air-stream heat exchanger **436** and the fuel-tank heat exchanger **448** to maintain a temperature of the high temperature heat source **416** below a predetermined heat load temperature.

(110) In some embodiments, the controller **468** may be configured to determine an amount of heat transfer to the fuel in the first fuel tank **444** and operate the plurality of valves **454**, **455**, **456**, **457**, **458** to selectively vary the flow of fluid in the cooling loop **420** through the air-stream heat exchanger **436** and the fuel-tank heat exchanger **448** in response to the amount of heat transferred being different from a predetermined heat rejection schedule.

(111) In some embodiments, the controller **468** is configured to determine a temperature of the fluid in the cooling loop **420** upstream of the accumulator **438** included in the cooling loop **420**. The controller **468** may operate the valves, **455**, **457** to vary the flow of fluid in the cooling loop **420** through the air-stream heat exchanger **436** in response to the temperature of the fluid in the cooling loop **420** upstream of the accumulator **438** measured by the sensor **467** being different than a sensor target temperature.

(112) In the illustrative embodiment, the controller **468** is configured to operate the plurality of valves **454**, **455**, **456**, **457**, **458** to allow the fluid to flow through the fuel-tank heat exchanger **448** based on inputs from the control system **426**. The plurality of valves **454**, **456**, **458** allow or block fluid from flowing through the fuel-tank heat exchanger **448** in response to the controller **468** receiving different signals. The modulating valves **455**, **457** control the flow of fluid through the air-stream heat exchanger **436**. The controller **468** may also be configured to operate the expansion valves **440**, **441** and the compressor **442** to allow the fluid to flow through the expansion valves **440**, **441** and the compressor **442**.

(113) The plurality of valves **454**, **456**, **458** allow fluid to flow through the fuel-tank heat exchanger **448** if the controller **468** receives a first signal and/or a second signal. The first signal is indicative of a temperature of the fuel in the first fuel tank **444** being equal to or above a predetermined maximum allowable tank temperature of the first fuel tank **444**, i.e. the maximum allowable temperature of the fuel in the first fuel tank **444** at that time of the mission. The second signal is indicative of a temperature of the cooling fluid directly upstream of the fuel-tank heat exchanger **448** being below the predetermined maximum allowable tank temperature, i.e. the maximum allowable temperature of the fuel in the first fuel tank **444** at that time of the mission.

(114) The predetermined maximum allowable tank temperature, or the target temperature for the first fuel tank **444**, is the maximum allowable temperature of the fuel in the first fuel tank **444** at that specific time of the mission. In the illustrative embodiment, the controller **468** is configured to determine the changes of the predetermined maximum allowable tank temperature throughout the mission.

(115) The maximum allowable fuel temperature for the engine **414** may also change throughout the

mission. Therefore, the predetermined maximum allowable fuel temperature is the maximum allowable temperature of the fuel supplied to the engine **414** at that time of the mission. In the illustrative embodiment, the controller **468** is configured to determine the changes of the predetermined maximum allowable fuel temperature throughout the mission.

(116) In the illustrative embodiment, the controller **468** may be configured determine the amount of heat to add to or remove from the first fuel tank **444** based on the predetermined maximum allowable fuel temperature at different times over the course of the mission and control the plurality of valves **454**, **456**, **458** accordingly. Therefore, the plurality of valves **454**, **456**, **458** may allow fluid to flow through the fuel-tank heat exchanger **448** based on the predetermined maximum allowable fuel temperature at different times over the course of the mission.

(117) In the illustrative embodiment, the controller **468** is configured to determine the changes of the predetermined maximum allowable tank temperature throughout the mission. In some instances, the predetermined maximum allowable tank temperature may be greater than the predetermined maximum allowable fuel temperature.

(118) In some embodiments, the plurality of valves **454**, **456**, **458** may allow fluid to flow through the fuel-tank heat exchanger **448** if the controller **468** receives a third signal. The third signal may be indicative of the air temperature outside the aircraft. In other words, the third signal may be indicative of the thermal energy system **12** being a predetermined altitude above ground where the air temperature is relatively cool. In other embodiments, the air temperature at ground level may be cool enough to provide cooling to the fuel tanks **444**, **446**.

(119) In the illustrative embodiments, the controller **468** is configured to operate the plurality of valves **454**, **456**, **458** to allow the fluid to flow through the fuel-tank heat exchanger **448** in response to the controller **468** receiving a fourth signal. The fourth signal is indicative of a take-off mode of the aircraft **10**.

(120) The engine-fuel unit **450** includes a mix valve **474**, conduits **476**, **478**, and the engine-fuel heat exchanger **479** as shown in FIGS. **6** and **7**. The first conduit **476** is in fluid communication between the mix valve **474** and the second fuel tank **446** to deliver fuel having a first temperature to the mix valve **474**. The second conduit **478** is in fluid communication between the mix valve **474** and the second fuel tank **446**. The engine-fuel heat exchanger **479** is in thermal communication with the second conduit **478** to cause the second conduit **478** to deliver fuel having a second temperature to the mix valve **474**.

(121) During use of the thermal energy system **412** in the aircraft **10**, the fluid in the fluid conduit **430** is pumped from an accumulator **438** included in the cooling loop **420** to the heat source **416** as indicated by arrow **482** and to the expansion valves **440**, **441** as indicated by arrow **481** in FIGS. **6** and **7**. The pump **432** moves the fluid through the fluid conduit **430** to transfer heat from the heat source **416** to the fluid to cool the heat source **416**. The pump **432** also moves the fluid through the expansion valve **440** as indicated by arrow **481** in FIGS. **6** and **7**. The fluid is expanded through the expansion valve **440** such that the fluid has a temperature lower than desired operating temperature of the battery **428**, in order to remove heat from the battery **428**.

(122) Moving from the heat source **416**, the flow of fluid in the fluid conduit **430** moves through the engine-fuel heat exchanger **479** of the engine-fuel unit **450** as indicated by arrow **483** in FIGS. **6** and **7**. The fluid in the fluid conduit **430** is in fluid communication with the engine-fuel heat exchanger **479** to transfer heat between the fluid to the fuel flowing and the engine **414**.

(123) Simultaneously, as the fluid exits the expansion valve **440**, the fluid in the fluid conduit **430** moves across the low temperature heat loads **428** as indicated by arrow **494** as shown in FIGS. **6** and **7**. As the fluid moves across the low temperature heat loads **428**, the low temperature heat loads **428** transfers heat to the fluid to cause the fluid to vaporize. The vapor then flows to the compressor **442** as indicated by arrow **496**. The compressor **442** compresses the vapor in the fluid conduit **430** to the pressure that is in the conduit **430**, and in the process increases the fluids temperature, in order for the fluid to be cooled by the engine-fuel heat exchanger **479** and/or the

air-stream heat exchanger **436**.

(124) The fluid from the compressor **442** and the heat source **416** then flows to the engine-fuel heat exchanger **479** as indicated by arrow **483** in FIGS. **6** and **7**. The fluid in the fluid conduit **430** may reject at least some heat to the fuel flowing through the engine-fuel heat exchanger **479** to cool the fluid in the cooling loop **420**. The mix valve **474** varies the flow of fuel from the fluid conduits **476**, **478** to adjust the temperature of the fuel provided to the engine **414**.

(125) As the fluid in the fluid conduit **430** exits the engine-fuel unit **450**, some or all of the fluid flow may be passed to the air-stream heat exchanger **436** as indicated by arrow **484** as shown in FIGS. **6** and **7**. The fluid flows through the air-stream heat exchanger **436** to transfer additional heat from the fluid in the cooling loop **420** to the air passing through the duct **434** of the cooler **422**.

(126) As the fluid in the fluid conduit **430** exits the air-stream heat exchanger **436** as indicated by arrow **486**, the valve **454** controls the flow of fluid to the thermal-storage fuel system **424**. The valve **454** is configured to selectively cause the flow of fluid to bypass the fuel-tank heat exchanger **448** based at least in part on the temperature measured by the sensors **462**, **463**, **464**, **466**, **467** as discussed above and as discussed in the other embodiments. If the controller **468** detects the first signal and/or the second signal, the valve **454** allows fluid to flow through the fuel-tank heat exchanger **448** as indicated by arrow **487** as shown in FIG. **6**.

(127) In the illustrative embodiment, as the valve **454** allows fluid in the fluid conduit **430** to flow to the fuel-tank heat exchanger **448**, the valve **454** partially or fully blocks flow from bypassing the fuel-tank heat exchanger **448** as suggested by the dotted line **492** in FIG. **6**. As such, little to no fluid is flowing in the fluid conduit **430** in the region of the dotted line **492** in FIG. **6**.

(128) Conversely, the valve **454** blocks the flow of the fluid through the fuel-tank heat exchanger **448** based on the temperature measured by the sensors **462**, **463**, **464**, **466**, **467** as indicated by the dotted arrow **487** shown in FIG. **7**. If the valve **454** partially or fully blocks the fluid in the fluid conduit **430** from flowing to the fuel-tank heat exchanger **448**, the rest or all of the fluid flows to the accumulator **438** as indicated by the arrow **492** in FIG. **7**.

(129) The valve **456** is connected to the fluid conduit **430** downstream of the expansion valve **441** and the valve **454** as shown in FIGS. **6** and **7**. The valve **456** is configured to selectively control the flow of fluid from the expansion valve **441** and the air-stream heat exchanger **436** based at least in part on the temperature measured by the sensors **462**, **464**, **466** as discussed above.

(130) If the controller **468** detects the first and second signals, the valve **456** is configured to allow all or some of the fluid in the cooling loop **420** exiting the air-stream heat exchanger **436** to flow through the fuel-tank heat exchanger **448** as indicated by arrow **488** in FIG. **6**.

(131) Conversely, if the temperature of the fuel in the first fuel tank **444** is equal to or above the predetermined maximum allowable tank temperature and the temperature of the air outside the aircraft **10** is relatively too hot, the valve **456** may fully, or partially block the fluid in the cooling loop **420** exiting the air-stream heat exchanger **436** through the fuel-tank heat exchanger **448**.

Similarly, if the temperature of the fuel in the first fuel tank **444** is equal to or above the predetermined maximum allowable fuel temperature and the temperature of the fluid exiting the air-stream heat exchanger **436** is above the predetermined maximum tank temperature, the valve **456** may fully, or partially block fluid in the cooling loop **420** to flow through the fuel-tank heat exchanger **448**.

(132) In such instances, the expansion valve **441** may be allowed to provide fluid to cool the fuel in the first fuel tank **444**. The fluid from the expansion valve **441** is able to further cool the fuel in the first fuel tank **444** than what is possible with the air-stream heat exchanger **436**. In some instances, the air flowing through the air-stream heat exchanger **436** may be relatively high when the aircraft **10** and thus the thermal energy system **12** is below the predetermined altitude.

(133) In some embodiments, if the temperature of the fuel in the first fuel tank **444** is equal to or above the predetermined maximum allowable tank temperature and the air flowing through the air-stream heat exchanger **436** is relatively high, the expansion valve **441** allows the fluid from the

expansion valve **441** to flow through the fuel-tank heat exchanger **448** as indicated by the arrow **488** in FIG. 7. Likewise, if the temperature of the fuel in the first fuel tank **444** is equal to or above the predetermined maximum allowable fuel temperature and the temperature of the fluid exiting the air-stream heat exchanger **436** is above the predetermined maximum tank temperature, the valve **456** allows fluid in the cooling loop **420** from the expansion valve **441** to flow through the fuel-tank heat exchanger **448**.

(134) The fluid in the cooling loop **420** from the expansion valve **441** is at a lower temperature than the temperature of the fluid at sensor **466**, which may allow the fluid to remove heat from the fuel in the first fuel tank **444** at a faster rate and/or to a lower temperature than if cooled with fluid coming from heat exchanger **436**. The fluid in the cooling loop **420** from the expansion valve **441** flowing through the fuel-tank heat exchanger **448** is vaporized and compressed. The third valve **458** controls the fluid in the cooling loop **420** as it exits the fuel-tank heat exchanger **448**.

(135) If the second valve **456** allowed the flow of fluid from the air-stream heat exchanger **436**, the valve **458** directs the flow of fluid exiting the fuel-tank heat exchanger **448** (indicated by arrow **490**) to return to the accumulator **438** as indicated by arrow **493** in FIG. 6. As such, the valve **458** fully blocks the flow of fluid to the compressor **442** as indicated by dotted arrow **497** in FIG. 6. Furthermore, a check valve **443** may be added downstream of the compressor **442**, to prevent liquid from entering the compressor **442** when the compressor **442** is not in operation.

(136) Conversely, if the second valve **456** allows the flow of fluid from the expansion valve **441**, the valve **458** allows the flow of the fluid exiting the fuel-tank heat exchanger **448** to flow to the compressor **442** as indicated by arrow **497** in FIG. 7. As such, the valve **458** fully blocks the flow of fluid to the accumulator **438** as indicated by dotted line **493** in FIG. 7.

(137) A method of cooling heat loads on the aircraft **10** may include several steps. The method includes conducting the fluid through the cooling loop **420** to transfer heat between the first heat source **416** and the fluid in the cooling loop **420** and conducting the fluid through the cooling loop **420** to transfer heat between the second heat source **428** and the fluid in the cooling loop **420**. The method also includes conducting the fluid in the cooling loop **420** to the air-stream heat exchanger **436** to transfer heat between the fluid in the cooling loop **420** and the air passing through the duct **434** and conducting the fluid in the cooling loop **420** to the fuel-tank heat exchanger **448** to transfer heat between the fluid in the cooling loop **420** and fuel stored in the first fuel tank **444**.

(138) In the illustrative embodiment, the method further includes selectively varying a flow of the fluid in the cooling loop **420** through the air-stream heat exchanger **436** and a flow of the fluid through the fuel-tank heat exchanger **448**. Varying the flow of the fluid through the air-stream heat exchanger **436** and the fuel-tank heat exchanger **448** maintains a temperature of the first heat source **416** below the predetermined heat load temperature.

(139) In the illustrative embodiment, the method further comprises conducting the fluid in the cooling loop **420** to the engine-fuel unit **450**. The fluid in the cooling loop **420** is conducted to the engine-fuel unit **450** before the fluid is conducted to the air-stream heat exchanger **436** to transfer heat between the fluid in the cooling loop **420** and the fuel in the engine-fuel unit **450**.

(140) In the illustrative embodiment, the method further includes varying a flow of fuel through the engine-fuel heat exchanger **479**. The flow of fuel is varied through the engine-fuel heat exchanger **479** to deliver fuel to the engine **414** at the engine-fuel unit predetermined threshold temperature.

(141) Another embodiment of a thermal energy system **512** in accordance with the present disclosure is shown in FIG. 8. The thermal energy system **512** is substantially similar to the thermal energy system **412** shown in FIGS. 6 and 7 and described herein. Accordingly, similar reference numbers in the **500** series indicate features that are common between the thermal energy system **512** and the thermal energy system **412**. The description of the thermal energy system **412** is incorporated by reference to apply to the thermal energy system **512**, except in instances when it conflicts with the specific description and the drawings of the thermal energy system **512**.

(142) The thermal energy system **512** includes a cooling loop **520**, a cooler **522**, a thermal-storage

fuel system **524**, a control system **526**, a first heat source **516**, and a second heat source **528** as shown in FIG. **8**. The cooling loop **520** has a fluid conduit **530** and a pump **532** configured to move fluid through the fluid conduit **530** to transfer heat from the heat sources **516**, **528** to the fluid in the fluid conduit **530** thereby cooling the heat sources **516**, **528**. The cooler **522** includes a duct **534** configured to receive air from the atmosphere **18** and an air-stream heat exchanger **536** located in the duct **534** to transfer heat between the fluid in the cooling loop **520** and the air passing through the duct **534**. The thermal-storage fuel system **524** is configured to transfer heat between the fluid in the cooling loop **520** and fuel stored in the thermal-storage fuel system **524**. The control system **526** is configured to control the flow of fluid through the fluid conduit **530**.

(143) The control system **526** includes a plurality of valves **554**, **555**, **556**, **557**, **558**, **560**, **561**, sensors **562**, **563**, **564**, **566**, **567**, and a controller **568** as shown in FIG. **8**. The valves **554**, **555**, **556**, **557**, **558**, **560**, **561** are connected to the fluid conduit **530** and configured to selectively control the flow of fluid through the cooling loop **520**. The sensors **562**, **563**, **564**, **566**, **567** are configured to measure the temperature of the fuel in the fuel tanks **544**, **546** and the temperature of the fluid in the fluid conduit **530**. The controller **568** is configured to operate the plurality of valves **554**, **555**, **556**, **557**, **558**, **560**, **561** to allow the fluid to flow through the fluid conduit **530** based at least in part on the information measured by the sensors **562**, **563**, **564**, **566**, **567**.

(144) The controller **568** is configured to manage the cooling of the heat loads **516** as well as the use of the thermal-storage fuel system **524**. The controller **568** is configured to direct the valves **554**, **555**, **556**, **557**, **558**, **560**, **561** to vary the flow of the fluid in the cooling loop **520**. The controller **568** is also configured to control the fuel pump flow rate of the fuel pump **547** included in the thermal-storage fuel system **524** through the fuel-tank heat exchanger **548**. The controller **568** controls the valves **554**, **555**, **556**, **557**, **560**, **561** and the fuel pump **547** so as to maintain the fluid in the cooling loop **520** at a predetermined temperature.

(145) The plurality of valves **554**, **555**, **556**, **557**, **558**, **560**, **561** includes a first valve **554**, a second valve **556**, a third valve **558**, a fourth valve **560**, a fifth valve **561**, and the modulating valves **555**, **557** as shown in FIG. **8**. The first valve **554** is connected to the fluid conduit **530** downstream of the air-stream heat exchanger **536** so as to selectively cause the flow of fluid to bypass the fuel-tank heat exchanger **548**. The second valve **556** is connected to the fluid conduit **530** downstream of the first valve **554** and an expansion valve **541** included in the cooling loop **520** so as to selectively control the flow of fluid through the fuel-tank heat exchanger **548**. The third valve **558** is connected to the fluid conduit **530** downstream of the fuel-tank heat exchanger **548** so as to selectively control the flow of fluid to the compressor **542** or back to the accumulator **538**. The fourth valve **560** is connected to the fluid conduit **530** upstream of a compressor **542** connected to the fluid conduit **530** so as to selectively control the flow of fluid through the compressor **542**. The fifth valve **561** is connected to the fluid conduit **530** upstream of an engine-fuel unit **550** included in the system **524** to vary the flow through the engine-fuel unit **550**. The valves **555**, **557** are configured to vary the flow rate of the fluid in the cooling loop **520** through the air-stream heat exchanger **536**.

(146) The control system **526** is configured to selectively vary a flow of the fluid in the cooling loop **520** through the air-stream heat exchanger **536** and the fuel-tank heat exchanger **548** to maintain a temperature of the high temperature heat source **516** below a predetermined heat load temperature.

(147) In some embodiments, the controller **568** may be configured to determine an amount of heat transfer to the fuel in the first fuel tank **544** and operate the plurality of valves **554**, **555**, **556**, **557**, **558** to selectively vary the flow of fluid in the cooling loop **520** through the air-stream heat exchanger **536** and the fuel-tank heat exchanger **548** in response to the amount of heat transfer being different from a predetermined heat rejection schedule.

(148) In some embodiments, the controller **568** is configured to determine a temperature of the fluid in the cooling loop **520** upstream of the accumulator **538** included in the cooling loop **520**. The controller **568** may operate the valves, **555**, **557** to vary the flow of fluid in the cooling loop

520 through the air-stream heat exchanger 536 in response to the temperature of the fluid in the cooling loop 520 upstream of the accumulator 538 measured by the sensor 567 being different than a sensor target temperature.

(149) In the illustrative embodiment, the second expansion valve 541 is arranged upstream of the thermal-storage fuel system 524. The compressor 542 is arranged downstream of the low temperature heat loads 528. In the illustrative embodiment, the expansion valves 540, 541 may act as control valves to control the flow of the fluid across the low temperature heat loads 528. In other embodiments, the expansion valves 540, 541, in conjunction with the compressor 542, may be configured to reduce the pressure of the fluid in the cooling loop 520 between the expansion valves 540, 541 and the compressor 542.

(150) The thermal-storage fuel system 524 further includes the engine-fuel unit 550 as shown in FIG. 8. Both the first fuel tank 544 and second fuel tank 546 are configured to store fuel therein. The engine-fuel unit 550 is configured to receive fuel from the second fuel tank 546 and deliver the fuel to the engine 514. In the illustrative embodiment, the engine-fuel unit 550 is in thermal communication with the cooling loop 520 to transfer heat between the fluid in the cooling loop 520 and the fuel in the engine-fuel unit 550.

(151) The engine-fuel unit 550 includes a mix valve 574, conduits 576, 578, and an engine-fuel heat exchanger 579 as shown in FIG. 8. The first conduit 576 is in fluid communication between the mix valve 574 and the second fuel tank 546 to deliver fuel having a first temperature to the mix valve 574. The second conduit 578 is in fluid communication between the mix valve 574 and the second fuel tank 546. The engine-fuel heat exchanger 579 is in thermal communication with the second conduit 578 to cause the second conduit 578 to deliver fuel having a second temperature to the mix valve 574.

(152) In the illustrative embodiment, the controller 568 is configured to operate the plurality of valves 554, 555, 556, 557, 558, 560, 561 to control the fluid to flow through the fuel-tank heat exchanger 548, the engine-fuel unit 550, and the compressor 542 based on inputs from the control system 526. Similar to the embodiments of FIGS. 6 and 7, the plurality of valves 554, 556, 558 allow fluid to flow through the fuel-tank heat exchanger 548 if the controller 568 receives the appropriate signals. In the illustrative embodiment, the controller 568 is also configured to operate the fourth valve 560 to control the fluid through the compressor 542 based on inputs from the control system 526.

(153) During use of the thermal energy system 512 in the aircraft 10, the fluid in the fluid conduit 530 is pumped from an accumulator 538 included in the cooling loop 520 to the heat source 516 as indicated by arrow 582 and to the expansion valves 540, 541 as indicated by arrow 581 in FIG. 8. The pump 532 moves the fluid through the fluid conduit 530 to transfer heat from the heat source 516 to the fluid to cool the heat source 516. The pump 532 also moves the fluid through the expansion valve 540 to expand the fluid such that the fluid has a relatively low temperature compared to the low temperature heat loads 528.

(154) As the fluid exits the expansion valve 540, the fluid in the fluid conduit 530 moves across the low temperature heat loads 528 as indicated by arrow 594 as shown in FIG. 8. As the fluid moves across the low temperature heat loads 528, the low temperature heat loads 528 transfer heat to the fluid to cause the fluid to be a vapor. The vapor then flows to the compressor 542 as indicated by arrow 596. The compressor 542 compresses the vapor in the fluid conduit 530 to be cooled by the thermal-storage fuel system 524 or the engine-fuel heat exchanger 579 and/or the air-stream heat exchanger 536 in some embodiments.

(155) However, on colder days, thermal lift may not be needed, as the air temperature may be low enough to sufficiently cool the low temperature heat loads 528. In this case, the accumulator 538 may be operated at a lower temperature without the need of the compressor 542 to lower the pressure and temperature in the low temperature heat loads 528. As such, the expansion valve 540 may operate as flow control valves and the fluid passing through the low temperature heat loads

528 may be vaporized. The resulting vaporized or partially vaporized fluid may then bypass the compressor **542** as indicated by the dotted arrow **598** as shown in FIG. **8**, which indicates no fluid flow through the compressor **542**.

(156) In some embodiments, the valve **560** may be eliminated and the fluid may be passed through the compressor **542** that is not running. However, the valve **560** may be included in the event that stopping the compressor **542** would not permit the flow of fluid through the compressor **542**. Furthermore, a check valve **543** may be added downstream of the compressor **542**, to prevent liquid from entering the compressor **542** when the compressor **542** is not in operation.

(157) Instead, the controller **568** is configured to direct the valve **560** to direct the vaporized or partially vaporized fluid back to the accumulator **538**. The valve **560** therefore blocks the flow of fluid to the compressor **542** as indicated by the dotted arrow **598**.

(158) In the illustrative embodiment, the cooling loop **520** further includes an orifice **531** as shown in FIG. **8**. The orifice **531** is connected to the fluid conduit **530** upstream of the heat source **516** to increase the pump exit pressure of the pump **532**. Increasing the pump exit pressure forces the flow through the low temperature heat loads **528**. Depending on the pressure drops through the various components **516**, **548**, **528**, etc., the orifice **531** may be removed. In some embodiments, the controller **568** may be configured to control the orifice **531**.

(159) The present disclosure relates to an aircraft **10** with a thermal energy system **12** to cool low temperature heat loads of the electrical components or other engine or aircraft oil/hydraulic heat loads. Hybrid electric thermal energy systems may rely on ram-air heat exchangers for rejecting the waste heat produced by power electronic devices or electric machines such as generators and motors.

(160) In the illustrative embodiments, the gas turbine engine **14** is mounted to the outside of the aircraft **10**. In other embodiments, the gas turbine engine **14**, **214**, **314**, **414**, **514** may be inside of the aircraft **10**. The engine **14**, **214**, **314**, **414**, **514** may be in the fuselage where electric power is produced. This electric power may then be transferred to an electric fan. The electric fan may be in other parts of the aircraft **10**, such as the tail, or there may be may electric fans distributed on the wings of the aircraft **10**.

(161) The thermal energy system **12** may use single-phase liquid coolants or two-phase coolants. For two-phase coolants, the cooling loop **20**, **220**, **320**, **420**, **520** may represent a two-phase pump loop (TPPL). The use of the ram air may be driven by the low temperature requirements of the devices being cooled. This low temperature difference between the component being cooled and the ram air temperature causes several challenges. In other embodiments, a fan stream duct may be used.

(162) The controller **68**, **268**, **468**, **568** may be configured to direct the valves **54**, **254**, **454**, **456**, **458**, **554**, **556**, **558** to selectively control the flow of fluid in the cooling loop **20**, **220**, **420**, **520** through the fuel-tank heat exchanger **48**, **248**, **448**, **548** to maximize fuel savings of the aircraft **10** based on the type of day (i.e. hot day or cold day). The controller **68**, **268**, **468**, **568** may direct the valves **54**, **254**, **454**, **456**, **458**, **554**, **556**, **558** based on the temperature along the flight path, while keeping the high temperature heat sources **16**, **216**, **316**, **416**, **516** and/or fuel below a predetermined threshold.

(163) First, when the aircraft **10** is sitting on the ground or taxiing, there is little to no ram pressure to force air through the air-stream heat exchanger **36** to cool the coolant. As such, a typical aircraft system may include a blower to force air through the heat rejection heat exchanger. The blower and additional ducting adds weight and volume to the thermal management system. The system of the present disclosure does not include a blower.

(164) Second, the highest heat loads typically occur at take-off, when the ram air is quite close in temperature to the devices to be cooled. Therefore very large mass flow rates of air may be needed to carry the heat away. Furthermore, at start of take-off, the ram pressure is relatively low. The large airflow rates and low ram pressure rise may cause typical heat rejection heat exchangers and

ram air ducts to be relatively large in size. The ducts and heat exchangers of the present disclosure may be small in size to reduce weight.

(165) Conversely, at cruise altitude, the ram air temperature will have significantly decreased and the ram pressure difference on the heat exchanger will have increased. As a result, a much smaller heat exchanger could theoretically be used on typical aircraft. However, because the heat rejection heat exchanger may have been sized for hot day take-off, the typical aircraft may experience a large drag penalty.

(166) To eliminate the blower and reduce the size of the duct and heat rejection heat exchanger, the thermal energy system **12, 212, 312, 412, 512, 612** of the present disclosure includes a thermal-storage fuel system **24, 224, 324, 424, 524, 624** that increases the efficiency of the aircraft **10**.

(167) The thermal-storage fuel system **24, 224, 324, 424, 524** enables a smaller ram air cooler **22, 222, 322, 422, 522** and eliminates the blower. The thermal-storage fuel system **24, 224, 324, 424, 524** uses the available on-board fuel as a heat sink for all or a portion of the electrical loads produced early in the mission while at low altitude (e.g. idle, taxi, take-off).

(168) Then, at altitude, the heat stored in the thermal-storage fuel system **24, 224, 324, 424, 524** may be rejected to the much colder ambient air. As a result, the air-stream heat exchanger **36, 236, 336, 436, 536** and the duct **34, 234, 334, 434, 534** may be sized for a condition that is not as challenging as the worst case condition of rejecting high heat loads to hot air temperatures with little ram pressure, resulting in smaller, lighter, and more efficient thermal system **12, 212, 312, 412, 512**.

(169) The heat transferred to the fluid in the cooling loop **20, 220, 320, 420, 520** is removed from the fluid using several different methods. In some embodiments, the engine-fuel unit **250, 350, 450, 550** rejects heat to the fuel that is being passed to the engine **214, 314, 414, 514** to be burned. In some embodiments, the ram-air heat rejection heat exchanger **36, 236, 336, 436, 536** rejects heat to the air flowing through the duct **34, 234, 334, 434, 534**. In some embodiments, the thermal-storage fuel system **24, 224, 324, 424, 524** stores the heat from the fluid in the fuel in the first fuel tank **44, 244, 344, 444, 544**. Any combination of these heat management features may be used. Each of these heat rejection methods helps remove heat from the heat source **16, 216, 316, 416, 516** out of the system **12, 212, 312, 412, 512** at different points in the mission or during different flight conditions.

(170) In some conventional thermal energy systems, the fuel is transferred from the tank, heated by the heat loads of the system, possibly cooled, and then returned to the tank. Moving the fuel around the system may be hazardous and cause safety issues.

(171) In this disclosure, the thermal-storage fuel system **24, 224, 324, 424, 524** keeps the fuel in close proximity to the respective fuel tank **44, 244, 344, 444, 544** and eliminates the running of fuel around the aircraft **10**, which reduces the risk of fire. In the illustrative embodiments of FIGS. 2-5, the system **12, 312** uses a single-phase coolant (e.g. water based coolant, oil, etc.). However, in the embodiments of FIGS. 2-4, the system **12** may also use a two-phase coolant.

(172) In the illustrative embodiments of FIGS. 6-8, the system **212, 412, 512** uses a two-phase coolant. In the case of two-phase coolant, the pumping power and pump size may also be decreased relative to the pumping of single phase coolant.

(173) In the illustrative embodiments, the thermal-storage fuel system **24, 224, 324, 424, 524** includes a dedicated thermal management system (TMS) tank **44, 244, 344, 444, 544** configured to enable a portion of the fuel to be taken to a temperature higher than what the engine **14, 214, 314, 414, 514** may safely receive. In this way, the fuel heat storage capacity is increased. The fuel may later be cooled to a lower temperature to be able to be safely supplied to the engine **14, 214, 314, 414, 514** later in the mission.

(174) In the illustrative embodiments, the fuel-tank heat exchanger **48, 248, 348, 448, 548** is shown to be located in the TMS tank **44, 244, 344, 444, 544**. In other embodiments, the fuel-tank heat exchanger **48, 248, 348, 448, 548** may be located adjacent to or outside of the TMS tank **44, 244,**

344, 444, 544. The fuel-tank heat exchanger **48, 248, 348, 448, 548** may contact the outside of the TMS fuel tank **44, 244, 344, 444, 544** in some embodiments.

(175) In other embodiments, the fuel-tank heat exchanger **48, 248, 348, 448, 548** is located adjacent to or in close proximity to the thermal-storage fuel system **24, 224, 324, 424, 524**. In other embodiments, the fuel-tank heat exchanger **48, 248, 348, 448, 548** may be located in another suitable location and fluidly connected to the fluid conduit **30, 230, 330, 430, 530** of the cooling loop **20, 220, 320, 420, 520**.

(176) The thermal energy system **12, 212, 312, 412, 512** extends a simple two-phase pump loop architecture to move and dissipate heat to various stations along the TPPL. The thermal-storage fuel system **24, 224, 324, 424, 524** stores the heat load on board and the air-stream heat exchanger **36, 236, 336, 436, 536** dissipates the heat to air when it can be done more efficiently. This reduces the size and drag of the condenser on the aircraft **10**.

(177) In the illustrative embodiments of FIGS. **6-8**, the thermal energy system **412, 512** further includes a compressor **442, 542** to enable cooling the low temperature heat loads **428, 528** to a lower temperature. The compressor **442, 542** may provide the capability of cooling the low temperature heat loads on a hot day. The compressor **442, 542** may also allow the other heat loads to operate at higher temperatures, thus making greater use of the thermal-storage fuel system **424, 524**. In other embodiments, the low temperature heat loads **428, 528** may be another low temperature device, such as avionics, directed energy weapons, etc.

(178) At the engine-fuel unit **250, 350, 450, 550**, the coolant rejects heat to the fuel provided to the engine **214, 314, 414, 514** to be burned. In order to maximize the use of the fuel heat sink, the fuel entering the engine **214, 314, 414, 514** is increased in temperature to its maximum allowable value or maximum allowable fuel temperature.

(179) To ensure the fuel provided to the engine **214, 314, 414, 514** is at the maximum allowable fuel temperature, the fuel stream is split prior to the engine-fuel heat exchanger **279, 479, 579** and only a portion of the fuel passes through the engine-fuel heat exchanger **279, 479, 579**. A mix valve **274, 474, 574** configured to sense the temperature of the fuel exiting the mix valve **274, 474, 574** is used to manage the temperature of the fuel entering the engine **214, 414, 514**.

(180) The mix valve **274, 474, 574** varies the amount of flow that passes through the engine-fuel heat exchanger **279, 479, 579**. If the coolant temperature is just hot enough to heat the fuel to the maximum allowable fuel temperature, the mix valve **274, 474, 574** will pass all of the fuel through the engine-fuel heat exchanger **279, 479, 579**.

(181) If the fuel coming from the second fuel tank **246, 446, 546** is already at the maximum allowable fuel temperature, then no fuel will pass through the engine-fuel heat exchanger **279, 479, 579**. In some embodiments, the split may be somewhere between these two extremes.

(182) The thermal-storage fuel system **24, 224, 324, 424, 524** includes the aircraft fuel tanks **44, 46, 244, 246, 344, 346, 444, 446, 544, 546** where some of the heat loads in the coolant are rejected. The fuel tanks **44, 46, 244, 246, 344, 346, 444, 446, 544, 546** are divided into at least two functional tanks (each functional tank can be made up of multiple tanks). The second fuel tank **46, 246, 346, 446, 546** may be very similar to fuel tanks typically used on aircraft **10**.

(183) However, the first fuel tank **44, 244, 344, 444, 544** may be a dedicated fuel tank(s) configured to isolate a portion of the aircraft fuel. In doing so, this fuel may be heated to predetermined threshold temperature or maximum allowable tank temperature. This temperature may be higher than a maximum allowable tank temperature of the second fuel tank **46, 246, 346, 446, 546** and the maximum allowable fuel temperature. In some embodiments, the maximum temperature in the first fuel tank **44, 244, 344, 444, 544** may be limited to the highest temperature of the coolant in the cooling loop **20, 220, 320, 420, 520**.

(184) However, if the coolant maximum temperature is too high, an upper limit may be put on the maximum allowable tank temperature of the first fuel tank **44, 244, 344, 444, 544**. Later in the mission, the fuel in the first fuel tank **44, 244, 344, 444, 544** may be cooled to or below maximum

allowable tank temperature of the first fuel tank **44, 244, 344, 444, 544**, so that the engine **14, 214, 314, 414, 514**, may safely receive the fuel. In some embodiments, a fuel pump may be used to pump fuel through the fuel-tank heat exchanger **48, 248, 348, 448, 548**.

(185) In some embodiments, the heat loads may be low enough that the first fuel tank **44, 244, 344, 444, 544** may not need to be heated higher than maximum allowable tank temperature of the second fuel tank **46, 246, 346, 446, 546**. In this case, the dedicated first fuel tank **44, 244, 344, 444, 544** may not be needed.

(186) In other embodiments, the fuel may be mixed between the second fuel tank **46, 246, 346, 446, 546** and the first fuel tank **44, 244, 344, 444, 544** until the second fuel tank **46, 246, 346, 446, 546** reaches maximum allowable tank temperature of the second fuel tank **46, 246, 346, 446, 546**. In doing so the time before the first fuel tank **44, 244, 344, 444, 544** reaches the maximum allowable tank temperature of the first fuel tank **44, 244, 344, 444, 544**, is extended. This allows the aircraft **10** to reach a higher altitude (and hence colder air) before all cooling is supplied by the air-stream heat exchanger **36, 236, 336, 436, 536**.

(187) In the illustrative embodiment, the fuel-tank heat exchanger **48, 248, 348, 448, 548** is located in the first fuel tank **44, 244, 344, 444, 544**. However, in other embodiments, the fuel-tank heat exchanger **48, 248, 348, 448, 548** may be located in another suitable location to heat the fuel in the first fuel tank **44, 244, 344, 444, 544**.

(188) The ram air cooler **22** includes a variable NACA inlet **33**. The variable NACA inlet **33** will reduce the amount of drag created from the duct **34**.

(189) Furthermore, on very cold days when the fuel is very cold, it may be advantageous to minimize heat rejection via the ram air cooler **22, 222, 322, 422, 522** to minimize drag and to dump the heat to the fuel. In this situation, the variable inlet **33** may be closed off, forcing the remaining coolant heat to be rejected to the first fuel tank **44, 244, 344, 444, 544**. Therefore, some of the waste heat may be retained in the engine cycle, improving system efficiency.

(190) Unlike other ram air coolers, the blower and blower duct/door are removed from the ram air cooler **22, 222, 322, 422, 522** of the present disclosure. This first fuel tank **44, 244, 344, 444, 544** is instead used as thermal energy storage when at ground idle or when taxiing.

(191) Because the air stream may not be needed until later in the mission, when at a higher altitude, the fan stream in an electric propulsor may be sufficiently cool to use as the air heat sink. If this is the case, a dedicated ram air duct may not be needed.

(192) To help with understanding the operation of this system **12, 212, 312, 412, 512**, the different modes of operation through a mission will be described. The mission to be described is for a very hot day.

(193) At ground idle, taxi, take-off, or start of climb, very little heat may be rejected through the air-stream heat exchanger **36, 236, 336, 436, 536**. As much heat as possible may be transferred to the fuel through the engine-fuel heat exchanger **279, 479, 579**, most of the heat may be rejected through the fuel-tank heat exchanger **48, 248, 348, 448, 548** and stored in the fuel in the first fuel tank **44, 244, 344, 444, 544**.

(194) As the aircraft **10** climbs and increases in speed, the air in the atmosphere **18** may become colder and the pressure difference across the air-stream heat exchanger **36, 236, 336, 436, 536** will increase. Therefore, even more heat from the coolant may be rejected through the air-stream heat exchanger **36, 236, 336, 436, 536** and less heat will be rejected to the first fuel tank **44, 244, 344, 444, 544**. At some point, all the heat may be rejected from the engine-fuel unit **250, 350, 450, 550** and the ram air cooler **22, 222, 322, 422, 522**.

(195) At the end of the climb, as the aircraft begins to cruise, the coolant exiting the ram air cooler **22, 222, 322, 422, 522** may have lost more heat than is being added to the coolant from the heat source **16, 216, 316, 416, 516**. At this point, the coolant from the air-stream heat exchanger **36, 236, 336, 436, 536** flowing to the fuel-tank heat exchanger **48, 248, 348, 448, 548** may begin to absorb heat from the fuel in the first fuel tank **44, 244, 344, 444, 544**.

(196) During the cruise phase of the mission, the first fuel tank **44, 244, 344, 444, 544** may be chilled below maximum allowable fuel temperature so that the fuel may be burned by the engine **14, 214, 314, 414, 514**. It may also be useful to significantly lower the fuel temperature to a very low temperature. In other words, the first fuel tank **44, 244, 344, 444, 544** may be deeply cooled before landing.

(197) As the aircraft **10** descends to lower altitudes, the air through the duct **34, 234, 334, 434, 534** may begin to heat up again. During descent, heat source **16, 216, 316, 416, 516** may be significantly less than during climb. However, a mission optimized ram air duct **34, 234, 334, 434, 534** may not be able to provide all of the needed cooling during the full final portion of the mission.

(198) As a result, the first fuel tank **44, 244, 344, 444, 544** may be used as heat storage once again. At this point, there is only a portion of the earlier fuel remaining. This remaining fuel may be made up of fuel yet to be burned and fuel reserves. However, if the fuel had been previously deeply cooled to a low temperature, the limited fuel mass may still provide significant thermal storage because it is so cold relative to maximum allowable fuel temperature.

(199) A variation on the mission profile may be for cold days. On cold days, the fuel may provide a very large thermal heat sink. In this case, one may close off the ram air cooler **22, 222, 322, 422, 522** and reject more of the heat into the very cold fuel in the first fuel tank **44, 244, 344, 444, 544**. Furthermore, because the second fuel tank **46, 246, 346, 446, 546** is so cold, rejecting heat into the second fuel tank **46, 246, 346, 446, 546** may be advantageous. In this way, electrical energy losses may be returned to the fuel heat sink and minimize energy that may be dumped overboard. Furthermore, by avoiding use of the ram air cooler **22, 222, 322, 422, 522**, ram drag affects may be minimized.

(200) In the illustrative embodiment of FIGS. **6** and **7**, the system **412** includes batteries **428**, which operate at relatively low temperatures. The illustrative embodiment, may add thermal lift capability to provide cooling capability at lower temperatures for the batteries **428**.

(201) In such illustrative embodiments, the coolant may be operated in two phases (i.e. liquid and vapor). The coolant may be a refrigerant that is chosen for optimal efficiency in the temperature operating range of the components being cooled and the heat sink that the heat is rejected to.

(202) The thermal energy system **412** includes expansion valves **440, 441** and a compressor **442**. The compressor **442** lowers the pressure downstream of the expansion valves **440, 441** indicated by arrows **496, 485**. As the liquid refrigerant is expanded across the expansion valve **440**, the refrigerant temperature is lowered and is able to keep the batteries **428** at a lower temperature than the other heat loads. As the refrigerant absorbs heat in the batteries **428**, the coolant is vaporized and the compressor **442** then compresses the refrigerant to return it to the coolant to be cooled.

(203) On cold days, thermal lift may not be needed, because the air temperature may be low enough to sufficiently cool the batteries **428, 528** using a two-phase pump loop, without thermal lift. In that case, the expansion valve **440** will have a lower pressure drop and will be used to control the flow rate to the batteries **428, 528**. As the coolant is passed through the low temperature heat loads **428**, the coolant may be partially or fully vaporized.

(204) In the illustrative embodiment of FIG. **8**, the resulting two-phase flow may then bypass the compressor **542** and be returned to the accumulator **538**. This approach would likely use an orifice **531** just before the heat source **516** to increase the pump exit pressure in order to force flow through the batteries **528**. The fluid flowing through the batteries **528** is then directed back to the accumulator **538** by the valve **560**. In this mode of operation, the full system may be behaving as a TPPL.

(205) Alternatively, the valve **560** may allow the fluid in line **596** to flow through line **598** to the compressor. The fluid may then flow through compressor **542** to eventually pass through line **583** to the engine-fuel heat exchanger **579**. The compressor **542** may have a bypass line to allow fluid to flow in this way if the compressor **542** is of a positive displacement design.

(206) In the illustrative embodiments of FIGS. 6 and 7, the valves **454, 456, 458** controls the flow of the fluid to and from the fuel-tank heat exchanger **448**. In some embodiments, the low temperature expanded refrigerant may be supplied to the fuel-tank heat exchanger **448** as show in FIG. 7. This may enable dropping the temperature of the fuel in the first fuel tank **444** to a temperature lower than the air temperature in the cooler **422**. This may be useful on very hot day operations to drive the fuel temperature to a very low temperature in preparation for using it as a heat sink at the end of the mission.

(207) The control system **26, 226, 426, 526** is configured to control the flow of fluid in the cooling loop **20, 220, 420, 520** to improve the overall performance of the aircraft **10**. To do so, the first step may be to maintain acceptable operating temperatures on the devices being cooled. The second may be to minimize fuel burn over the entire mission.

(208) In doing this, the controller **68, 268, 468, 568** may chose when or how to reject the heat load **16, 216, 416, 516, 616** from the cooling loop **20, 220, 420, 520**. Hence, the controller **68, 268, 468, 568** may be configured to put as much heat into the fuel tanks **44, 46, 244, 246, 344, 346, 444, 446, 544, 546**, without ever encountering a situation where the devices are hotter than some upper predetermined threshold limit.

(209) On a hot day, on a short duration flight, the use of the first fuel tank **44, 244, 344, 444, 544** may be minimized so that during descent and/or taxiing, the first tank **44, 244, 344, 444, 544** is cold enough to provide cooling.

(210) However, on a longer flight, perhaps more heat may be added to the first tank **44, 244, 344, 444, 544**. During the long flight, the first tank **44, 244, 344, 444, 544** may start being cooled with a higher initial temperature because there is sufficient time to cool the tank **44, 244, 344, 444, 544** back down to the temperature required during descent and taxiing.

(211) In the next extreme, on a cold day flight, the fuel tanks **44, 46, 244, 246, 344, 346, 444, 446, 544, 546** may be heated to minimize drag from the ram duct **34, 234, 334, 434, 534** and to minimize waste heat from being dumped overboard of the aircraft **10**. As such, the controller **68, 268, 468, 568** may be configured to look at the mission profile and weather along the flight path to minimize fuel burn while still maintaining acceptable temperatures on the components.

(212) In the illustrative embodiments, the sensors **62, 63, 64, 66, 67, 262, 263, 264, 266, 267, 462, 463, 464, 466, 467, 562, 563, 564, 566, 567** are shown in respective locations to measure the temperature of the fuel or fluid in the system **12, 212, 312, 412, 512**. In other embodiments, the sensors **62, 63, 64, 66, 67, 262, 263, 264, 266, 267, 462, 463, 464, 466, 467, 562, 563, 564, 566, 567** may be located along other points along the fluid conduit **30, 230, 430, 530, 630** that are suitable to measure the temperature of the fluid in the cooling loop **20, 220, 420, 520**. In other embodiments, the system **12, 212, 412, 512** may include more sensors.

(213) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A thermal energy system for use with an aircraft, the thermal energy system comprising: a heat source, a cooling loop having a fluid conduit and a pump configured to move a fluid through the fluid conduit to transfer heat from the heat source to the fluid to cool the heat source, a cooler that includes a duct configured to conduct a flow of air through the duct and an air-stream heat exchanger located in the duct and in thermal communication with the fluid in the cooling loop to transfer heat between the cooling loop and the flow of air conducted through the duct, a thermal-storage fuel system that includes a TMS fuel tank configured to store a first fuel therein at a first temperature, an aircraft fuel tank independent of the TMS fuel tank configured to store a second

fuel at a second temperature that is different from the first temperature of the first fuel in the TMS fuel tank, and a fuel-tank heat exchanger in thermal communication with the fluid in the cooling loop and configured to transfer heat between the fluid in the cooling loop and the first fuel from the TMS fuel tank, an engine-fuel unit configured to receive the second fuel from the aircraft fuel tank and deliver the second fuel from the aircraft fuel tank to an engine, and a controller connected with one or more valves, a valve system, and one or more sensors; wherein the engine-fuel unit includes the valve system and an engine-fuel heat exchanger, the valve system is in fluid communication with the aircraft fuel tank, the engine-fuel heat exchanger is in thermal communication with the valve system and the cooling loop and is configured to transfer heat between the fluid in the cooling loop and the second fuel in the engine-fuel unit delivered from the aircraft fuel tank, wherein the controller is configured to vary a flow of the second fuel from the aircraft fuel tank through the engine-fuel heat exchanger via the valve system to deliver the second fuel from the aircraft fuel tank to the engine at a third temperature that is less than or equal to an engine-fuel unit predetermined threshold temperature, wherein the controller is configured to vary a flow of the fluid in the cooling loop through the fuel-tank heat exchanger via the one or more valves to heat the first temperature of the first fuel in the TMS fuel tank to a first fuel tank predetermined threshold temperature which is greater than the engine-fuel unit predetermined threshold temperature.

2. The thermal energy system of claim 1, wherein the engine-fuel unit predetermined threshold temperature is equal to a maximum allowable fuel temperature.

3. The thermal energy system of claim 1, wherein the aircraft fuel tank is in fluid communication with the TMS fuel tank, and wherein the one or more valves further includes an inlet valve and an outlet valve that are each fluidly connected to the TMS fuel tank and the aircraft fuel tank and are each configured to respectively control a flow of the first fuel and a flow of the second fuel between the TMS fuel tank and the aircraft fuel tank.

4. The thermal energy system of claim 3, wherein the controller is configured to selectively vary at least one of the flow of the second fuel from the aircraft fuel tank through the inlet valve to the TMS fuel tank and the flow of the first fuel from the TMS fuel tank through the outlet valve to the aircraft fuel tank if the first temperature of the first fuel in the TMS fuel tank is approaching the first fuel tank predetermined threshold temperature.

5. The thermal energy system of claim 1, wherein the cooling loop is structured to conduct the fluid heated from the heat source first through the engine-fuel heat exchanger, second through the air-stream heat exchanger after through the engine-fuel heat exchanger, and third through the fuel-tank heat exchanger after through the air-stream heat exchanger.

6. The thermal energy system of claim 1, wherein the valve system includes a mix valve, a first conduit in fluid communication between the mix valve and the aircraft fuel tank to deliver a first portion of the second fuel from the aircraft fuel tank at the second temperature to the mix valve, and a second conduit in fluid communication with the mix valve, the engine-fuel heat exchanger, and the aircraft fuel tank, wherein the engine-fuel heat exchanger is in thermal communication with the second conduit to transfer heat between the fluid in the cooling loop and a second portion of the second fuel from the aircraft fuel tank to cause the second conduit to deliver the second portion of the second fuel from the aircraft fuel tank at a fourth temperature to the mix valve, and wherein the controller is configured to vary a first flow rate of the first portion of the second fuel having the second temperature from the first conduit and a second flow rate of the second portion of the second fuel having the fourth temperature from the second conduit via the mix valve to provide a mixed stream of the second fuel having the third temperature that is less than or equal to the engine-fuel unit predetermined threshold temperature.

7. The thermal energy system of claim 1, wherein the one or more valves includes a bypass valve connected to the fluid conduit, and the controller is configured to selectively cause at least a portion of a flow of the fluid in the cooling loop to bypass the fuel-tank heat exchanger via the bypass valve in order to vary the flow of the fluid passing through the fuel-tank heat exchanger.

8. The thermal energy system of claim 7, wherein the one or more valves further includes modulating valves connected to the fluid conduit, and the controller is configured to vary a flow of the fluid in the cooling loop through the air-stream heat exchanger via the modulating valves.
9. The thermal energy system of claim 1, wherein the controller is configured to selectively vary a first flow of the fluid in the cooling loop through the air-stream heat exchanger and a second flow of the fluid in the cooling loop through the fuel-tank heat exchanger via the one or more valves to maintain a temperature of the heat source below a predetermined heat load temperature, and wherein the controller is configured to vary the predetermined heat load temperature throughout a flight cycle of the aircraft.
10. The thermal energy system of claim 9, wherein the controller is configured to measure a heat transfer between the second flow of the fluid in the cooling loop through the fuel-tank heat exchanger and the first fuel in the TMS fuel tank via the one or more sensors, compare the measured heat transfer to a predetermined heat rejection schedule, and selectively vary the second flow of the fluid in the cooling loop through the fuel-tank heat exchanger via the one or more valves in response to the measured heat transfer being different from the predetermined heat rejection schedule.
11. The thermal energy system of claim 9, wherein the controller is configured to vary the first fuel tank predetermined threshold temperature throughout the flight cycle of the aircraft.
12. The thermal energy system of claim 1, wherein the cooler is a ram air cooler and the duct is configured to receive the flow of air from an atmosphere around the aircraft during forward movement of the aircraft relative to ground and is free of any air mover.
13. A method comprising providing a thermal energy system for use with an aircraft, the thermal energy system including: a cooling loop, a cooler, a thermal-storage fuel system, an engine-fuel unit, and a controller, the cooling loop having a fluid conduit and a pump configured to move a fluid through the fluid conduit, the cooler including a duct configured to conduct a flow of air through the duct and an air-stream heat exchanger located in the duct and in thermal communication with the fluid in the cooling loop to transfer heat between the cooling loop and the flow of air conducted through the duct, the thermal-storage fuel system including a TMS fuel tank configured to store a first fuel therein at a first temperature, an aircraft fuel tank in fluid communication with the TMS fuel tank and configured to store a second fuel at a second temperature that is different from the first temperature of the first fuel in the TMS fuel tank, a fuel-tank heat exchanger in thermal communication with the fluid in the cooling loop, and the engine-fuel unit in thermal communication with the cooling loop and configured to receive the second fuel from the aircraft fuel tank and deliver the second fuel from the aircraft fuel tank to an engine, the controller connected with one or more valves, a valve system, and one or more sensors; conducting the fluid through the cooling loop to transfer heat between a heat source and the fluid in the cooling loop, conducting the fluid in the cooling loop to the engine-fuel unit to transfer heat between the fluid in the cooling loop and at least a portion of the second fuel in the engine-fuel unit delivered from the aircraft fuel tank, conducting the fluid in the cooling loop to the air-stream heat exchanger to transfer heat between the fluid in the cooling loop and the flow of air passing through the duct, conducting the fluid in the cooling loop to the fuel-tank heat exchanger to transfer heat between the fluid in the cooling loop and the first fuel from the TMS fuel tank, wherein the engine-fuel unit includes the valve system and an engine-fuel heat exchanger, the valve system is in fluid communication with the aircraft fuel tank, and the engine-fuel heat exchanger is in thermal communication with the valve system and the cooling loop and is configured to transfer heat between the fluid in the cooling loop and the at least a portion of the second fuel delivered from the aircraft fuel tank, wherein the controller is configured to vary a flow of the at least a portion of the second fuel from the aircraft fuel tank through the engine-fuel heat exchanger via the valve system to deliver the second fuel from the aircraft fuel tank to the engine at a third temperature less than or equal to an engine-fuel unit predetermined threshold temperature, and wherein the method further

comprises varying a flow of the fluid in the cooling loop through the fuel-tank heat exchanger to heat the first temperature of the first fuel in the TMS fuel tank to a first fuel tank predetermined threshold temperature which is greater than the engine-fuel unit predetermined threshold temperature.

14. The method of claim 13, further comprising selectively varying a flow of the fluid in the cooling loop through the air-stream heat exchanger and the flow of the fluid through the fuel-tank heat exchanger to maintain a temperature of the heat source below a predetermined heat load temperature and varying the predetermined heat load temperature throughout a flight cycle of the aircraft.

15. The method of claim 13, wherein the valve system includes a mix valve, a first conduit in fluid communication between the mix valve and the aircraft fuel tank to deliver a first portion of the second fuel from the aircraft fuel tank at the second temperature to the mix valve, and a second conduit in fluid communication with the mix valve, the engine-fuel heat exchanger, and the aircraft fuel tank, wherein the engine-fuel heat exchanger is in thermal communication with the second conduit to transfer heat between the fluid in the cooling loop and a second portion of the second fuel from the aircraft fuel tank to cause the second conduit to deliver the second portion of the second fuel from the aircraft fuel tank at a fourth temperature to the mix valve, and wherein the controller is configured to vary a first flow rate of the first portion of the second fuel from the first conduit and a second flow rate of the second portion of the second fuel from the second conduit via the mix valve to provide a mixed stream of the second fuel having the third temperature that is less than or equal to the engine-fuel unit predetermined threshold temperature.

16. The method of claim 13, wherein the one or more valves includes an inlet valve and an outlet valve that are each fluidly connected to the TMS fuel tank and the aircraft fuel tank and are each configured to respectively control a flow of the first fuel and a flow of the second fuel between the TMS fuel tank and the aircraft fuel tank, and wherein the method further comprises transferring an amount of the second fuel in the aircraft fuel tank through the inlet valve to the TMS fuel tank if the first temperature of the first fuel in the TMS fuel tank is approaching the first fuel tank predetermined threshold temperature.

17. The method of claim 13, wherein the one or more valves includes an inlet valve and an outlet valve that are each fluidly connected to the TMS fuel tank and the aircraft fuel tank and are each configured to respectively control a flow of the first fuel and a flow of the second fuel between the TMS fuel tank and the aircraft fuel tank, and wherein the method further comprises transferring an amount of the first fuel from the TMS fuel tank through the outlet valve to the aircraft fuel tank if the first temperature of the first fuel in the TMS fuel tank is approaching the first fuel tank predetermined threshold temperature.

18. The method of claim 13, further comprising measuring a heat transfer between the flow of the fluid in the cooling loop through the fuel-tank heat exchanger and the first fuel in the TMS fuel tank, comparing the measured heat transfer to a predetermined heat rejection schedule, and selectively varying the flow of the fluid in the cooling loop passing through the fuel-tank heat exchanger in response to the measured heat transfer being different from the predetermined heat rejection schedule.

19. The method of claim 13, further comprising measuring a fluid temperature of the fluid in the cooling loop upstream of an accumulator included in the cooling loop, comparing the measured fluid temperature to a sensor target temperature, and selectively varying a flow of the fluid in the cooling loop passing through the air-stream heat exchanger in response to the measured fluid temperature being different from the sensor target temperature.

20. The method of claim 13, wherein the engine-fuel unit predetermined threshold temperature is equal to a maximum allowable fuel temperature.
